

Multi-LED Visible Light Positioning System With LED Signal Selection or Combining

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Abstract—Light-emitting diode (LED) based visible light positioning (VLP) has been widely studied to provide high-accuracy and low-cost localization services. Existing single-LED VLP systems suffer from severe energy attenuation when the receiver is far from the LED. Even though multiple LEDs could be deployed, the fusion of their positioning information remains an open problem. In this paper, we propose a multi-LED indoor VLP system with LED signal selection or combining. The receiver contains several differently-oriented photodiodes (PDs) working cooperatively as an angle-of-arrival estimator, and each LED individually provides the receiver with a localization result. With these single-LED positioning results, we select the results with the minimum localization error, or combine the results according to their distributions. Closed-form positioning error expressions are derived for the proposed methods. Moreover, increasing the LED number is shown to reduce the positioning error, but the reduction is negligible when the LED number is sufficiently large. Finally, an experimental testbench is built to verify the feasibility of the proposed methods, and a millimeter-level accuracy has been achieved, which is better than that of the classic trilateration method.

Index Terms—Angle-of-arrival, equal gain combining, LED signal selection, maximum likelihood combining, visible light positioning

I. INTRODUCTION

HIGH-accuracy indoor positioning has aroused increasing attention of researchers due to the high demands of future communication and automation systems. Though global navigation satellite systems (GNSS) have been widely applied to outdoor positioning, they fail to provide accurate indoor localization services due to the satellite signal attenuation incurred by walls and multipath propagation [1]. As the supplements to GNSS, indoor radio-frequency (RF) positioning systems based on WiFi, Bluetooth, ZigBee, RF identification device, etc were proposed [2]. However, the RF indoor positioning systems generally achieve a low accuracy of meter-level due to the exploitation of low-frequency electromagnetic waves that propagate with notable diffractions and reflections.

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When high-frequency signals are exploited, ultra wideband (UWB) positioning can reduce the localization error to within a few centimeters [2], but it requires extra hardware and suffers from a high cost, while the reflectors must be sufficiently sparse.

Compared with the RF positioning methods, visible light positioning (VLP) is a promising candidate for the indoor localization due to its high accuracy and low cost [3], [4]. In VLP systems, photodiodes (PDs) or cameras are usually exploited to receive the signals transmitted from the light-emitting diodes (LEDs). VLP methods can be roughly classified into three groups: received signal strength (RSS), time-of-arrival (TOA) or time-difference-of-arrival (TDOA), and angle-of-arrival (AOA). The RSS method measures the exact received light powers to estimate the lengths of optical links, and then uses trilateration algorithm to locate the user [5]–[8]. Such a method is easy to deploy, but highly relies on the accurate transmitting and receiving models, and the exact values of the system parameters, such as LEDs' transmission powers, which limit the positioning accuracy of the RSS method. The TOA or TDOA method measures the travel time of the optical signal from the transmitter to the receiver, and then estimates the length of the optical link. Though TOA/TDOA-based systems were reported in [9]–[11], such systems were difficult to implement in practice due to the stringent requirement of time synchronization. Compared with the former two methods, the AOA method does not require a precise time synchronization and an accurate transmitting model, which is considered as the most promising VLP method [12]. AOA-based systems first estimate the incident light directions, and then localize the user based on the known coordinates of the light sources. The incident light directions can be determined with the light spot positions sensed by cameras [13]–[15] or the relative light powers received by PD arrays [16]–[18].

Since AOA is promising among the VLP methods, we have reviewed several AOA-based VLP systems in detail. Reference [19] proposed an AOA-RSS-based VLP system in 2014, in which a tilted-multiple-receiver structure was designed to receive the signals from an LED array. However, the proposed system in [19] lacked a closed-form expression for the positioning result, and the expected incidence angle gain difference of each receiver pair over the room needed to be pre-computed based on the LED position. After matching the incidence angle gain differences obtained from the received signals to the expected ones, the receiver could be finally localized. Therefore, the proposed method in [19] is inefficient

due to the time-consuming gain-difference calculation and matching. In 2019, Zhu named the structure of multiple differently-oriented PDs as an AOA estimator, and used it to estimate the incident light direction of an LED [20]. In this work, Zhu derived an optimal structure of the AOA estimator to minimize the AOA estimation error and proposed a two-dimensional (2D) VLP system. However, since only one LED was exploited in [20], the positioning results were accurate only near the LED. Exploiting multiple LEDs in the VLP system might be a feasible solution to expand the area of accurate localization. In [21], Cao proposed an approach to select the LED results and derived a closed-form error expression for the dual-LED cases. However, the selection scheme is not the optimal and has not been compared with the combining schemes. In 2022, Zhu proposed another AOA VLP system based on the eigenvalue decomposition of the covariance matrix of light direction vectors [22]. With multiple LEDs, ref. [22] could estimate the position and the rotation of the user simultaneously. In [23], an image sensor (IS) was exploited to detect the signals from several beacon LEDs to localize the user. However, the IS's bandwidth is generally not sufficient to distinguish LEDs' identity (ID) codes. Although rolling shutter techniques can be applied to distinguish IDs of kHz with CMOS cameras [24], they require the LEDs to be sufficiently large to extract the stripes correctly, and would cause jelly effect for the ISs equipped on the mobile users. This indicates that the ISs or cameras are relatively unsuitable to be applied to the scenario of multiple LEDs. Except for the cameras or tilted PDs, ref. [25] designed a quadrant PD angular diversity aperture to estimate the light direction passing through the aperture. Though various VLP systems were investigated in the aforementioned works, they lack explicit methods to fuse the positioning signals when multiple LEDs are deployed. Moreover, closed-form positioning error expressions should be derived for the fused positioning results to conveniently evaluate the localization performance.

In this paper, we propose approaches to select or combine the VLP results for an indoor 2D-VLP system, where N LEDs are deployed. In this VLP system, an AOA estimator consisting of M differently-oriented PDs is localized. We first localize the AOA estimator with each single LED based on the method in [20]¹, and thus obtain N single-LED positioning results. Then, we process these positioning results with different approaches to generate several new location estimations of the AOA estimator. The main contributions of this work are summarized as follows:

- Closed-form positioning error expression is derived for the single-LED 2D-VLP method in [20].
- The LED-select method is proposed to choose the LED of the minimum positioning error.
- Equal gain combining (EGC) and maximum likelihood combining (MLC) are exploited to fuse the N single-LED positioning results.

- Closed-form positioning error expressions are derived for the proposed methods.
- The relationship between the LED number and the positioning error is investigated.
- An experimental platform is built to test the proposed positioning methods.

Since the combining methods are designed for localization results, they can also fuse the results from multi-LED or other sensors, and support distributed processing², which have better versatilities than conventional VLP methods.

This paper is organized as follows. A system model is given in Section II. Section III shows the error of the single-LED positioning. In Section IV, the LED-select method is proposed and its positioning error is derived. Section V demonstrates the localization result combining methods and the derivations of their positioning errors. The simulation results are shown in Section VI, and the experimental results are demonstrated in Section VII. Section VIII draws concluding remarks.

In this paper, scalars are denoted by *italics* letters, vectors are expressed by lowercase bold ones, and matrices are represented in uppercase bold. $\|\mathbf{x}\|$ denotes the 2-norm of the vector $\mathbf{x}$. $E(\mathbf{X})$ denotes the expectation of the matrix $\mathbf{X}$. $\mathbf{I}_n$ denotes an $n \times n$ identity matrix. $\mathbf{0}_n$ denotes an $n \times 1$ zero vector. $\text{tr}(\mathbf{X})$ denotes the trace of the matrix $\mathbf{X}$. $|\mathbf{X}|$ denotes the determinant of the matrix $\mathbf{X}$. $\min(x_1, x_2, \dots, x_N)$ and $\max(x_1, x_2, \dots, x_N)$ denote functions that output the minimum and maximum of $x_1, x_2, \dots, x_N$, respectively. $\sum_{i=1}^N x_i$ and $\prod_{i=1}^N x_i$ denote the sum and product of $x_1, x_2, \dots, x_N$, respectively. $\Pr(A)$ denotes the probability of the event A . $Q(x)$ denotes the Gaussian Q function. $\mathcal{N}(\mu, \sigma_n^2)$ denotes the Gaussian distribution of the expectation μ and the variance σ_n^2 .

II. SYSTEM MODEL

In Fig. 1, we demonstrate a VLP system, where N LEDs are on the ceiling and an AOA estimator [20] to be localized is on the ground. The AOA estimator contains M differently-oriented PDs, and the j th PD's normal vector is denoted by a 3×1 vector $\mathbf{v}_j$. The positions of the i th LED and the AOA estimator center are denoted by 3×1 vectors $\mathbf{t}_i$ and $\mathbf{a}$, respectively. Therefore, the length-normalized directional vector from the AOA estimator to the i th LED is expressed as $\mathbf{r}_i = (\mathbf{t}_i - \mathbf{a})/\|\mathbf{t}_i - \mathbf{a}\|$. Since the signals of the LEDs can be distinguished by the AOA estimator with time division multiplexing (TDM) [16] or frequency division multiplexing (FDM) [26] scheme, we can individually perform a 2D localization of the AOA estimator based on each LED.

¹The concept of selecting an LED or combining the results from different LEDs was not mentioned in [20]. Therefore, though the same kind of the receiver is exploited, this work has different main contributions compared with [20].

²The raw data from the sensors could be processed by micro units to intermediate results, and then transmitted to the main processor for combining. This could reduce the burden on the main processor.

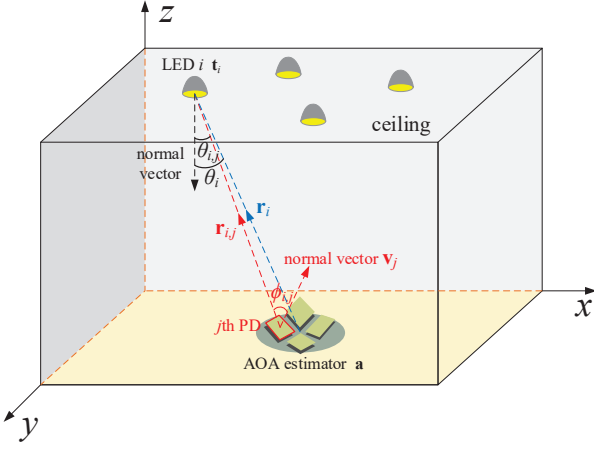


Fig. 1. Demonstration of a VLP system. N LEDs are set on the ceiling, each of which can be individually exploited to localize the AOA estimator on the ground. The positions of the i th LED and the AOA estimator are denoted by $\mathbf{t}_i$ and $\mathbf{a}$, respectively.

A. Received Signal of the AOA Estimator

According to [5, eq. (1)], the optical power signal received by the j th PD from the i th LED in Fig. 1 is expressed as

$$\begin{aligned}\hat{s}_{i,j} &= s_{i,j} + n_{i,j} \\ &= P_t \frac{(m+1)\alpha}{2\pi d_{i,j}^2} \cos^m \theta_{i,j} \cos \phi_{i,j} + n_{i,j} \\ &= P_t \frac{(m+1)\alpha}{2\pi d_{i,j}^2} \cos^m \theta_{i,j} \mathbf{v}_j^T \mathbf{r}_{i,j} + n_{i,j}\end{aligned}\quad (1)$$

where $\hat{s}_{i,j}$ and $s_{i,j}$ ($i = 1, \dots, N; j = 1, \dots, M$) denote the received noisy and noise-free signals of the j th PD from the i th LED, respectively; $n_{i,j}$ is the noise sample, which follows the Gaussian distribution of $\mathcal{N}(0, \sigma_n^2)$ [27]–[29]³; P_t denotes the transmission power of the LED; m is the Lambertian order of the LED; α denotes the active area of the PD; $d_{i,j}$ is the distance between the i th LED and the j th PD; $\theta_{i,j}$ denotes the angle between the i th LED's normal vector and the corresponding optical link; $\mathbf{r}_{i,j}$ denotes the length-normalized directional vector from the j th PD to the i th LED; $\phi_{i,j}$ denotes the angle between $\mathbf{v}_j$ and $\mathbf{r}_{i,j}$. Since the PDs are sufficiently small, they can be placed with negligible spacings in Fig. 1 [20], and thus they are assumed to be located at the AOA estimator center. Therefore, the variables $d_{i,j}$, $\theta_{i,j}$ and $\mathbf{r}_{i,j}$ can be assumed independent with the subscript j , and (1) is transformed as

$$\hat{s}_{i,j} = P_t \underbrace{\frac{(m+1)\alpha}{2\pi d_i^2} \cos^m \theta_i \mathbf{v}_j^T \mathbf{r}_i}_{p_{\max,i}} + n_{i,j} \quad (2)$$

where $d_i = \|\mathbf{t}_i - \mathbf{a}\|$; θ_i denotes the angle between the i th LED's normal vector and the directional vector $\mathbf{a} - \mathbf{t}_i$ as shown

³According to our experimental results, the input-independent Gaussian noise model is a reasonable channel model for our laboratory environment. Besides, the input-dependent Gaussian noise model would make the error analysis be much more complicated and would not bring insights for readers. The related research contents of the input-dependent Gaussian noise model are reserved as our future working directions.

in Fig. 1; $p_{\max,i}$ denotes the received optical power when the PD in the AOA estimator is right towards the i th LED ($\phi_{i,j} = 0$), which is the maximum optical power received by the PD from the i th LED.

B. AOA Estimation and Error Analysis

According to [20, eq. (5)], the AOA estimator outputs the directional vector sample $\hat{\mathbf{r}}_i$ as

$$\hat{\mathbf{r}}_i = \frac{1}{p_{\max,i}} (\mathbf{V}_{\text{PD}}^T \mathbf{V}_{\text{PD}})^{-1} \mathbf{V}_{\text{PD}}^T \hat{\mathbf{s}}_i \quad (3)$$

where $\hat{\mathbf{r}}_i$ is the estimation of $\mathbf{r}_i$; $\mathbf{V}_{\text{PD}} = [\mathbf{v}_1, \dots, \mathbf{v}_M]^T$ is a $M \times 3$ matrix that contains the normal vectors of the PDs; $\hat{\mathbf{s}}_i = [\hat{s}_{i,1}, \dots, \hat{s}_{i,M}]^T$ is a $M \times 1$ vector that includes the received signals of the PDs, each item of which has been expressed in (2). Applying (2) to (3), we obtain

$$\hat{\mathbf{r}}_i = \mathbf{r}_i + \underbrace{\frac{1}{p_{\max,i}} (\mathbf{V}_{\text{PD}}^T \mathbf{V}_{\text{PD}})^{-1} \mathbf{V}_{\text{PD}}^T \mathbf{n}_i}_{\mathbf{n}_i^r} \quad (4)$$

where $\mathbf{n}_i = [n_{i,1}, \dots, n_{i,M}]^T$ contains the noise samples of the PDs; $\mathbf{n}_i^r$ denotes the estimation error of $\hat{\mathbf{r}}_i$. Since $n_{i,1}, \dots, n_{i,M}$ are independent with each other and follow the Gaussian distribution of $\mathcal{N}(0, \sigma_n^2)$, we have

$$E(n_{i,a} n_{i,b}) = \begin{cases} 0 & a \neq b, \\ \sigma_n^2 & a = b. \end{cases} \quad (5)$$

In this paper, we adopt the optimal $\mathbf{V}_{\text{PD}}$ derived in [20, eq. (20)] as

$$\mathbf{V}_{\text{PD}} = \sqrt{\frac{2}{3}} \begin{bmatrix} \cos \frac{1}{M} 2\pi & \sin \frac{1}{M} 2\pi & \frac{1}{\sqrt{2}} \\ \cos \frac{2}{M} 2\pi & \sin \frac{2}{M} 2\pi & \frac{1}{\sqrt{2}} \\ \vdots & \vdots & \vdots \\ \cos \frac{M}{M} 2\pi & \sin \frac{M}{M} 2\pi & \frac{1}{\sqrt{2}} \end{bmatrix}, \quad (6)$$

and thus $\mathbf{V}_{\text{PD}}^T \mathbf{V}_{\text{PD}} = (M/3) \mathbf{I}_3$. Therefore, $\mathbf{n}_i^r$ in (4) is calculated as

$$\mathbf{n}_i^r = \frac{3}{M p_{\max,i}} \mathbf{V}_{\text{PD}}^T \mathbf{n}_i, \quad (7)$$

whose covariance matrix is expressed as

$$E(\mathbf{n}_i^r \mathbf{n}_i^{rT}) = \frac{3\sigma_n^2}{M p_{\max,i}^2} \mathbf{I}_3. \quad (8)$$

This indicates that the entries of $\mathbf{n}_i^r$ are independent and identically distributed.

C. Single-LED Positioning

Based on the estimated directional vector $\hat{\mathbf{r}}_i$ in (3), we localize the AOA estimator with the positioning algorithm developed in [20, eq. (30)] as

$$\hat{\mathbf{a}}_i = f(\hat{\mathbf{r}}_i) = \mathbf{t}_i - \frac{h}{[0, 0, 1] \hat{\mathbf{r}}_i} \hat{\mathbf{r}}_i \quad (9)$$

where $\hat{\mathbf{a}}_i$ denotes the single-LED localization result with the i th LED; h denotes the height of the ceiling in Fig. 1, or equivalently, the z -axis element of $\mathbf{t}_i$ is h . Equation (9)

contains one matrix addition and two matrix multiplications. Substituting $\mathbf{t}_i = [x_i, y_i, h]^T$ and $\hat{\mathbf{r}}_i = [\hat{r}_{i,1}, \hat{r}_{i,2}, \hat{r}_{i,3}]^T$ into (9), we have

$$\hat{\mathbf{a}}_i = \begin{bmatrix} x_i \\ y_i \\ h \end{bmatrix} - \frac{h}{\hat{r}_{i,3}} \begin{bmatrix} \hat{r}_{i,1} \\ \hat{r}_{i,2} \\ \hat{r}_{i,3} \end{bmatrix} = \begin{bmatrix} x_i - \frac{\hat{r}_{i,1}}{\hat{r}_{i,3}}h \\ y_i - \frac{\hat{r}_{i,2}}{\hat{r}_{i,3}}h \\ 0 \end{bmatrix}. \quad (10)$$

Therefore, the z -axis element of $\hat{\mathbf{a}}_i$ always equals 0, which implies that (9) is exactly a 2D localization method for the AOA estimator on the ground. Substituting $\hat{\mathbf{r}}_i = \mathbf{r}_i$ into (9), we have

$$\begin{aligned} f(\mathbf{r}_i) &= \mathbf{t}_i - \frac{h}{[0, 0, 1] \frac{(\mathbf{t}_i - \mathbf{a})}{\|\mathbf{t}_i - \mathbf{a}\|}} \frac{\mathbf{t}_i - \mathbf{a}}{\|\mathbf{t}_i - \mathbf{a}\|} \\ &= \mathbf{t}_i - \frac{h(\mathbf{t}_i - \mathbf{a})}{[0, 0, 1](\mathbf{t}_i - \mathbf{a})} \\ &= \mathbf{t}_i - \frac{h(\mathbf{t}_i - \mathbf{a})}{h} \\ &= \mathbf{a} \end{aligned} \quad (11)$$

which indicates that the 2D VLP method in (9) can localize the AOA estimator without error when the estimated directional vector $\hat{\mathbf{r}}_i$ contains no error.

D. VLP With Multiple LEDs

With the LEDs in Fig. 1, we can obtain N single-LED localization results $\hat{\mathbf{a}}_i$ s based on (9), and how to fuse these results to improve the localization accuracy should be investigated. For this problem, we propose two solutions:

- Selecting one of $\hat{\mathbf{a}}_i$ s that has a minimum positioning error, which is expressed as

$$\hat{\mathbf{a}} = \{\hat{\mathbf{a}}_i | e_i = \min(e_1, \dots, e_N)\} \quad (12)$$

where e_i denotes the positioning error of the i th single-LED localization result.

- Combining $\hat{\mathbf{a}}_i$ s to yield a new localization result of the AOA estimator, which is expressed as

$$\hat{\mathbf{a}} = g_{\text{com}}(\hat{\mathbf{a}}_1, \dots, \hat{\mathbf{a}}_N) \quad (13)$$

where $g_{\text{com}}(\cdot)$ denotes the combining method.

The two solutions in (12) and (13) will be developed in Section IV and Section V, respectively.

III. ERROR OF SINGLE-LED POSITIONING

In this section, we aim to investigate the localization error of the signal-LED positioning in (9), whose closed-form expression is shown in **Theorem III.1**.

Theorem III.1 (Single-LED localization error). *The positioning error with the i th LED is expressed as*

$$e_i = \sqrt{\left(1 + \frac{d_i^2}{h^2}\right) \frac{12\pi^2 \sigma_n^2 d_i^{2m+6}}{M(m+1)^2 P_t^2 \alpha^2 h^{2m}}}. \quad (14)$$

Proof. Since the error vector $\mathbf{n}_i^r$ in (4) could be assumed as a tiny disturbance to $\mathbf{r}_i$ when $\sigma_n \rightarrow 0$, we apply Taylor series expansion to (9) at $\hat{\mathbf{r}}_i = \mathbf{r}_i$, and obtain

$$\begin{aligned} \hat{\mathbf{a}}_i &= f(\mathbf{r}_i) + \frac{\partial f(\hat{\mathbf{r}}_i)}{\partial \hat{\mathbf{r}}_i} \bigg|_{\hat{\mathbf{r}}_i = \mathbf{r}_i} (\hat{\mathbf{r}}_i - \mathbf{r}_i) + o(\hat{\mathbf{r}}_i - \mathbf{r}_i) \\ &\approx \mathbf{a} + \frac{\partial \hat{\mathbf{a}}_i}{\partial \hat{\mathbf{r}}_i} \bigg|_{\hat{\mathbf{r}}_i = \mathbf{r}_i} \mathbf{n}_i^r \end{aligned} \quad (15)$$

where $o(\hat{\mathbf{r}}_i - \mathbf{r}_i)$ denotes a vector function whose elements are the higher order small quantities of $\hat{\mathbf{r}}_i - \mathbf{r}_i$. If we define $\hat{\mathbf{a}}_i = [\hat{a}_{i,1}, \hat{a}_{i,2}, \hat{a}_{i,3}]^T$, $\partial \hat{\mathbf{a}}_i / \partial \hat{\mathbf{r}}_i$ in (15) is calculated as

$$\frac{\partial \hat{\mathbf{a}}_i}{\partial \hat{\mathbf{r}}_i} = \begin{bmatrix} \frac{\partial \hat{a}_{i,1}}{\partial \hat{r}_{i,1}} & \frac{\partial \hat{a}_{i,1}}{\partial \hat{r}_{i,2}} & \frac{\partial \hat{a}_{i,1}}{\partial \hat{r}_{i,3}} \\ \frac{\partial \hat{a}_{i,2}}{\partial \hat{r}_{i,1}} & \frac{\partial \hat{a}_{i,2}}{\partial \hat{r}_{i,2}} & \frac{\partial \hat{a}_{i,2}}{\partial \hat{r}_{i,3}} \\ \frac{\partial \hat{a}_{i,3}}{\partial \hat{r}_{i,1}} & \frac{\partial \hat{a}_{i,3}}{\partial \hat{r}_{i,2}} & \frac{\partial \hat{a}_{i,3}}{\partial \hat{r}_{i,3}} \end{bmatrix}. \quad (16)$$

According to (10), eq. (16) is calculated as

$$\frac{\partial \hat{\mathbf{a}}_i}{\partial \hat{\mathbf{r}}_i} = \begin{bmatrix} -\frac{h}{\hat{r}_{i,3}} & 0 & \frac{h\hat{r}_{i,1}}{\hat{r}_{i,3}^2} \\ 0 & -\frac{h}{\hat{r}_{i,3}} & \frac{h\hat{r}_{i,2}}{\hat{r}_{i,3}^2} \\ 0 & 0 & 0 \end{bmatrix}. \quad (17)$$

Applying (17) to (15), we have

$$\hat{\mathbf{a}}_i \approx \mathbf{a} + \underbrace{\begin{bmatrix} -\frac{h}{r_{i,3}} & 0 & \frac{hr_{i,1}}{r_{i,3}^2} \\ 0 & -\frac{h}{r_{i,3}} & \frac{hr_{i,2}}{r_{i,3}^2} \\ 0 & 0 & 0 \end{bmatrix}}_{\mathbf{A}_i} \mathbf{n}_i^r \quad (18)$$

where $\mathbf{r}_i = [r_{i,1}, r_{i,2}, r_{i,3}]^T$ and $r_{i,1}^2 + r_{i,2}^2 + r_{i,3}^2 = 1$; $\partial \hat{\mathbf{a}}_i / \partial \hat{\mathbf{r}}_i|_{\hat{\mathbf{r}}_i = \mathbf{r}_i}$ in (15) is renamed as $\mathbf{A}_i$ in (18) for convenience. Subtracting $\mathbf{a}$ from both sides of (18), we have

$$\hat{\mathbf{a}}_i - \mathbf{a} \approx \mathbf{A}_i \mathbf{n}_i^r \quad (19)$$

which denotes the localization error vector when the i th LED is exploited. Based on (7) and (19), we can further calculate the positioning mean square error (MSE) as

$$\begin{aligned} e_i^2 &:= \mathbb{E}[(\hat{\mathbf{a}}_i - \mathbf{a})^T (\hat{\mathbf{a}}_i - \mathbf{a})] \\ &= \frac{9}{M^2 p_{\text{max},i}^2} \mathbb{E}[(\mathbf{A}_i \mathbf{V}_{\text{PD}}^T \mathbf{n}_i)^T \mathbf{A}_i \mathbf{V}_{\text{PD}}^T \mathbf{n}_i] \\ &= \frac{9}{M^2 p_{\text{max},i}^2} \mathbb{E}[\mathbf{n}_i^T \mathbf{V}_{\text{PD}} \mathbf{A}_i^T \mathbf{A}_i \mathbf{V}_{\text{PD}}^T \mathbf{n}_i] \end{aligned} \quad (20)$$

where e_i represents the positioning error with the i th LED. According to (5) and [30, eq. (14)], e_i in (20) is calculated as

$$\begin{aligned} e_i &= \sqrt{\frac{9\sigma_n^2}{M^2 p_{\text{max},i}^2} \text{tr}(\mathbf{V}_{\text{PD}} \mathbf{A}_i^T \mathbf{A}_i \mathbf{V}_{\text{PD}}^T)} \\ &= \sqrt{\frac{9\sigma_n^2}{M^2 p_{\text{max},i}^2} \text{tr}(\mathbf{A}_i \mathbf{V}_{\text{PD}}^T \mathbf{V}_{\text{PD}} \mathbf{A}_i^T)} \\ &= \sqrt{\frac{3\sigma_n^2}{M p_{\text{max},i}^2} \text{tr}(\mathbf{A}_i \mathbf{A}_i^T)}. \end{aligned} \quad (21)$$

According to (18), $\text{tr}(\mathbf{A}_i \mathbf{A}_i^T)$ is obtained as

$$\begin{aligned} \text{tr}(\mathbf{A}_i \mathbf{A}_i^T) &= \frac{2h^2}{r_{i,3}^2} + \frac{r_{i,1}^2 + r_{i,2}^2}{r_{i,3}^4} h^2 \\ &= \frac{2h^2}{r_{i,3}^2} + \frac{1 - r_{i,3}^2}{r_{i,3}^4} h^2 \\ &= \frac{h^2}{r_{i,3}^2} + \frac{h^2}{r_{i,3}^4}. \end{aligned} \quad (22)$$

Based on (2) and (22), we calculate (21) as

$$e_i = \sqrt{\frac{3\sigma_n^2}{M} \left[P_t \frac{(m+1)\alpha}{2\pi d_i^2} \cos^m \theta_i \right]^{-2} \left(\frac{h^2}{r_{i,3}^2} + \frac{h^2}{r_{i,3}^4} \right)}. \quad (23)$$

Since $\mathbf{r}_i = (\mathbf{t}_i - \mathbf{a}) / \|\mathbf{t}_i - \mathbf{a}\|$, we have

$$r_{i,3} = \frac{h}{d_i}. \quad (24)$$

If the LEDs in Fig. 1 have the same normal vector $[0, 0, -1]^T$, we have $\cos \theta_i = h/d_i$, and thus e_i in (23) is simplified as

$$e_i = \sqrt{\left(1 + \frac{d_i^2}{h^2}\right) \frac{12\pi^2 \sigma_n^2 d_i^{2m+6}}{M(m+1)^2 P_t^2 \alpha^2 h^{2m}}}. \quad (25)$$

□

Equation (14) quantifies the error of the single-LED positioning, which circumvents the time-consuming Monte Carlo simulations.

IV. LED SELECTION FOR MULTI-LED VLP SYSTEMS

In this section, we aim to select one optimal single-LED localization result from $\hat{\mathbf{a}}_i$ s in (9) for the AOA estimator.

A. LED Selection Method

Based on (14), we find that the positioning error e_i declines as d_i decreases when other parameters, such as P_t and h , are fixed. This indicates that the closest LED to the AOA estimator contributes the localization result of the minimum error. As shown in (24), d_i and $r_{i,3}$ are inversely proportional, which implies that (12) could be simplified to $\hat{\mathbf{a}} = \{\hat{\mathbf{a}}_i | r_{i,3} = \max(r_{1,3}, \dots, r_{N,3})\}$. Since $r_{i,3}$ is unavailable, we should select the LED based on $\hat{r}_{i,3}$ in (10). Such a selection method is expressed as

$$\hat{\mathbf{a}}_{\text{sel}} = \{\hat{\mathbf{a}}_i | \hat{r}_{i,3} = \max(\hat{r}_{1,3}, \dots, \hat{r}_{N,3})\} \quad (26)$$

where $\hat{\mathbf{a}}_{\text{sel}}$ denotes the optimal localization result with a single LED for the AOA estimator. The main advantages of the LED selection method are summarized: 1) it has a low complexity because it only needs to compare the values of N scalars; 2) it does not rely on the statistics of the AOA samples.

B. Positioning Error of LED Selection

According to the idea of selection, if the single-LED localization result $\hat{\mathbf{a}}_i$ is selected, the corresponding positioning MSE of the single-LED method (i.e. e_i^2) is the positioning MSE of the LED-select method. If $\Pr(\hat{\mathbf{a}}_{\text{sel}} = \hat{\mathbf{a}}_i)$ denotes the probability that $\hat{\mathbf{a}}_i$ is selected, the positioning MSE of the LED-select method has the probability of $\Pr(\hat{\mathbf{a}}_{\text{opt}} = \hat{\mathbf{a}}_i)$ to be e_i^2 . Therefore, the positioning MSE of the LED selection method in (26) is expressed as

$$e_{\text{sel}}^2 = \sum_{i=1}^N \Pr(\hat{\mathbf{a}}_{\text{sel}} = \hat{\mathbf{a}}_i) e_i^2 \quad (27)$$

where e_{sel} represents the positioning error with the LED selection; e_i is the single-LED positioning error that has been derived in (14). Based on (26), $\Pr(\hat{\mathbf{a}}_{\text{sel}} = \hat{\mathbf{a}}_i)$ is expressed as

$$\Pr(\hat{\mathbf{a}}_{\text{sel}} = \hat{\mathbf{a}}_i) = \Pr[\hat{r}_{i,3} = \max(\hat{r}_{1,3}, \dots, \hat{r}_{N,3})]. \quad (28)$$

To calculate the error in (27), we need to investigate the distribution of $\hat{r}_{i,3}$ in (28), which is shown in **Theorem IV.1**.

Theorem IV.1 (Distribution of $\hat{r}_{i,3}$). $\hat{r}_{i,3}$ in (28) is the z -axis entry of the estimated directional vector from the AOA estimator to the i th LED, i.e. $\hat{\mathbf{r}}_i$. $\hat{r}_{i,3}$ follows a Gaussian distribution of $\mathcal{N}(r_{i,3}, \sigma_i^2)$, where

$$\sigma_i^2 = \frac{3\sigma_n^2}{Mp_{\max,i}^2}. \quad (29)$$

Proof. According to (3) and (6), we have

$$\begin{aligned} \hat{\mathbf{r}}_i &= [\hat{r}_{i,1}, \hat{r}_{i,2}, \hat{r}_{i,3}]^T \\ &= \frac{\sqrt{6}}{Mp_{\max,i}} \begin{bmatrix} \cos \frac{1}{M} 2\pi & \cdots & \cos \frac{M}{M} 2\pi \\ \sin \frac{1}{M} 2\pi & \cdots & \sin \frac{M}{M} 2\pi \\ \frac{1}{\sqrt{2}} & \cdots & \frac{1}{\sqrt{2}} \end{bmatrix} \begin{bmatrix} \hat{s}_{i,1} \\ \hat{s}_{i,2} \\ \vdots \\ \hat{s}_{i,M} \end{bmatrix} \\ &= \frac{\sqrt{6}}{Mp_{\max,i}} \begin{bmatrix} \cos \frac{1}{M} 2\pi & \cdots & \cos \frac{M}{M} 2\pi \\ \sin \frac{1}{M} 2\pi & \cdots & \sin \frac{M}{M} 2\pi \\ \frac{1}{\sqrt{2}} & \cdots & \frac{1}{\sqrt{2}} \end{bmatrix} \begin{bmatrix} s_{i,1} + n_{i,1} \\ s_{i,2} + n_{i,2} \\ \vdots \\ s_{i,M} + n_{i,M} \end{bmatrix}, \end{aligned} \quad (30)$$

and it can be given that

$$\hat{r}_{i,3} = \frac{\sqrt{3}}{Mp_{\max,i}} \sum_{j=1}^M (s_{i,j} + n_{i,j}). \quad (31)$$

Therefore, the mean and the variance of $\hat{r}_{i,3}$ are calculated as

$$\mu_i := \mathbb{E}(\hat{r}_{i,3}) = \frac{\sqrt{3}}{Mp_{\max,i}} \sum_{j=1}^M s_{i,j}, \quad (32)$$

and

$$\sigma_i^2 := \text{var}(\hat{r}_{i,3}) = \frac{3}{M^2 p_{\max,i}^2} \sum_{j=1}^M \mathbb{E}(n_{i,j}^2) = \frac{3\sigma_n^2}{Mp_{\max,i}^2}. \quad (33)$$

Substituting (2) and (6) into (32), we have

$$\begin{aligned}\mu_i &= \frac{\sqrt{3}}{M} \sum_{j=1}^M \mathbf{v}_j^T \mathbf{r}_i \\ &= \frac{\sqrt{3}}{M} \sum_{j=1}^M \sqrt{\frac{2}{3}} \left[\cos \frac{j}{M} 2\pi, \sin \frac{j}{M} 2\pi, \frac{1}{\sqrt{2}} \right] \mathbf{r}_i \\ &= \frac{\sqrt{2}}{M} \sum_{j=1}^M \left(r_{i,1} \cos \frac{j}{M} 2\pi + r_{i,2} \sin \frac{j}{M} 2\pi + \frac{r_{i,3}}{\sqrt{2}} \right) \\ &= r_{i,3}\end{aligned}\quad (34)$$

where $\sum_{j=1}^M \cos(j/M)2\pi = \sum_{j=1}^M \sin(j/M)2\pi = 0$. Since $n_{i,1}, \dots, n_{i,M}$ are Gaussian random variables, $\hat{r}_{i,3}$ follows a Gaussian distribution of $\mathcal{N}(r_{i,3}, \sigma_i^2)$. $\square$

Based on **Theorem IV.1**, $\Pr(\hat{\mathbf{a}}_{\text{sel}} = \hat{\mathbf{a}}_i)$ in (27) is calculated as (35) on the top of the next page, where $Q(x) = \int_x^\infty 1/\sqrt{2\pi} \exp(-t^2/2) dt$ is the Gaussian Q function. Based on (14) and (35), the positioning error of the LED selection in (27) can be calculated.

Theorem IV.2 (LED-select localization error when $N = 2$). When $N = 2$, e_{sel} in (27) is expressed as

$$e_{\text{sel}} = \sqrt{Q\left(\frac{r_{j,3} - r_{i,3}}{\sqrt{\sigma_i^2 + \sigma_j^2}}\right) e_i^2 + Q\left(\frac{r_{i,3} - r_{j,3}}{\sqrt{\sigma_i^2 + \sigma_j^2}}\right) e_j^2}. \quad (36)$$

Proof. When $N = 2$, eq. (35) is calculated as

$$\begin{aligned}\Pr(\hat{\mathbf{a}}_{\text{sel}} = \hat{\mathbf{a}}_i) &= \Pr(\hat{r}_{i,3} > \hat{r}_{j,3}) \\ &= \int_{-\infty}^{+\infty} \left[1 - Q\left(\frac{\hat{r}_{i,3} - r_{j,3}}{\sigma_j}\right) \right] \\ &\quad \times \frac{1}{\sqrt{2\pi}\sigma_i} \exp\left[-\frac{(\hat{r}_{i,3} - r_{i,3})^2}{2\sigma_i^2}\right] d\hat{r}_{i,3}\end{aligned}\quad (37)$$

According to [31, eq. (63)], eq. (37) can be calculated as

$$\Pr(\hat{\mathbf{a}}_{\text{sel}} = \hat{\mathbf{a}}_i) = Q\left(\frac{r_{j,3} - r_{i,3}}{\sqrt{\sigma_i^2 + \sigma_j^2}}\right). \quad (38)$$

Therefore, when $N = 2$, e_{sel} in (27) is expressed as

$$e_{\text{sel}} = \sqrt{Q\left(\frac{r_{j,3} - r_{i,3}}{\sqrt{\sigma_i^2 + \sigma_j^2}}\right) e_i^2 + Q\left(\frac{r_{i,3} - r_{j,3}}{\sqrt{\sigma_i^2 + \sigma_j^2}}\right) e_j^2}. \quad (39)$$

$\square$

Theorem IV.3 (Lower bound of LED-select localization error). The lower bound of e_{sel} in (27) is expressed as

$$e_{\text{lb}} = \sqrt{\sum_{i=1}^N \left[e_i^2 \prod_{j=1, j \neq i}^N Q\left(\frac{r_{j,3} - r_{i,3}}{\sqrt{\sigma_i^2 + \sigma_j^2}}\right) \right]}. \quad (40)$$

Proof. Equation (35) can be bounded as

$$\begin{aligned}\Pr(\hat{\mathbf{a}}_{\text{sel}} = \hat{\mathbf{a}}_i) &= \Pr[\hat{r}_{i,3} = \max(\hat{r}_{1,3}, \dots, \hat{r}_{N,3})] \\ &= \Pr(\hat{r}_{i,3} > \hat{r}_{1,3}, \dots, \hat{r}_{i,3} > \hat{r}_{N,3}) \\ &\geq \prod_{j=1, j \neq i}^N \Pr(\hat{r}_{i,3} > \hat{r}_{j,3}).\end{aligned}\quad (41)$$

The derivation of (41) and the condition of its tightness are demonstrated in Appendix A. According to (37)-(38), eq. (41) is expressed as

$$\begin{aligned}\Pr(\hat{\mathbf{a}}_{\text{sel}} = \hat{\mathbf{a}}_i) &\geq \prod_{j=1, j \neq i}^N \Pr(\hat{r}_{i,3} > \hat{r}_{j,3}) \\ &= \prod_{j=1, j \neq i}^N Q\left(\frac{r_{j,3} - r_{i,3}}{\sqrt{\sigma_i^2 + \sigma_j^2}}\right),\end{aligned}\quad (42)$$

and thus the lower bound of e_{sel} in (27) is calculated as

$$e_{\text{lb}} = \sqrt{\sum_{i=1}^N \left[e_i^2 \prod_{j=1, j \neq i}^N Q\left(\frac{r_{j,3} - r_{i,3}}{\sqrt{\sigma_i^2 + \sigma_j^2}}\right) \right]}. \quad (43)$$

$\square$

V. LOCALIZATION RESULT COMBINING FOR MULTI-LED VLP SYSTEMS

In Sec. IV, only the localization result of one LED is selected, and the information provided by other LEDs is abandoned. In this section, we aim to exploit the information of all the LEDs, and develop more accurate localization methods.

A. Equal Gain Combining

In RF wireless communication systems, multiple received signals can be averaged to balance their power gaps, which is known as EGC. This method can also be applied to VLP systems. To average the positioning errors of different LEDs, we can treat the localization results with EGC as

$$\hat{\mathbf{a}}_{\text{egc}} = \frac{1}{N} \sum_{i=1}^N \hat{\mathbf{a}}_i \quad (44)$$

where $\hat{\mathbf{a}}_{\text{egc}}$ is the localization result obtained with the EGC, which contains $N - 1$ matrix additions and one matrix multiplication; $\hat{\mathbf{a}}_i$ is the single-LED localization result obtained with i th LED as defined in (9). Therefore, the positioning MSE of the EGC is expressed as

$$\begin{aligned}e_{\text{egc}}^2 &= \mathbb{E}[(\hat{\mathbf{a}}_{\text{egc}} - \mathbf{a})^T (\hat{\mathbf{a}}_{\text{egc}} - \mathbf{a})] \\ &= \mathbb{E}\left[\left(\frac{1}{N} \sum_{i=1}^N \hat{\mathbf{a}}_i - \mathbf{a}\right)^T \left(\frac{1}{N} \sum_{i=1}^N \hat{\mathbf{a}}_i - \mathbf{a}\right)\right] \\ &= \frac{1}{N^2} \sum_{i=1}^N \mathbb{E}[(\hat{\mathbf{a}}_i - \mathbf{a})^T (\hat{\mathbf{a}}_i - \mathbf{a})]\end{aligned}\quad (45)$$

$$\begin{aligned}
\Pr(\hat{\mathbf{a}}_{\text{sel}} = \hat{\mathbf{a}}_i) &= \Pr[\hat{r}_{i,3} = \max(\hat{r}_{1,3}, \dots, \hat{r}_{N,3})] \\
&= \int_{-\infty}^{+\infty} \frac{1}{\sqrt{2\pi}\sigma_i} \exp\left[-\frac{(\hat{r}_{i,3} - r_{i,3})^2}{2\sigma_i^2}\right] \underbrace{\left\{ \prod_{j=1, j \neq i}^N \int_{-\infty}^{\hat{r}_{j,3}} \frac{1}{\sqrt{2\pi}\sigma_j} \exp\left[-\frac{(\hat{r}_{j,3} - r_{j,3})^2}{2\sigma_j^2}\right] d\hat{r}_{j,3} \right\}}_{\Pr(\hat{r}_{j,3} < \hat{r}_{i,3})} d\hat{r}_{i,3} \\
&= \int_{-\infty}^{+\infty} \frac{1}{\sqrt{2\pi}\sigma_i} \exp\left[-\frac{(\hat{r}_{i,3} - r_{i,3})^2}{2\sigma_i^2}\right] \left\{ \prod_{j=1, j \neq i}^N \left[1 - Q\left(\frac{\hat{r}_{i,3} - r_{j,3}}{\sigma_j}\right)\right] \right\} d\hat{r}_{i,3}.
\end{aligned} \tag{35}$$

where e_{egc} represents the positioning error with the EGC. Applying (14) to (45), we have

$$e_{\text{egc}} = \frac{1}{N} \sqrt{\sum_{i=1}^N e_i^2}. \tag{46}$$

This indicates that the EGC method can average the positioning errors of different LEDs. However, it should be noted that when one or some of the LEDs are far away from the AOA estimator, $\sum_{i=1}^N e_i^2$ in (46) would become large, and thus the localization result of the EGC would be deteriorated by these LEDs. The main advantages of the EGC method are summarized: 1) it does not rely on the statistics of the AOA samples; 2) it generally has higher accuracy than selection, except for the cases when some LED signals are too weak.

B. Maximum Likelihood Combining

The main idea of MLC is to statistically analyze multiple received signals, optimize the parameters to maximize a likelihood possibility, and then combine the received signals with certain weights to estimate the transmitted signals. Here, we apply this method to $\hat{\mathbf{a}}_i$ s to derive a more accurate positioning result of the AOA estimator. The MLC method requires the probability density function (PDF) of the single-LED localization result.

Theorem V.1 (PDF of single-LED localization result). *The PDF of the single-LED localization result with the i th LED is expressed as*

$$\begin{aligned}
\Pr(\hat{\mathbf{a}}_i^{2D} | \mathbf{a}^{2D}) \\
= \frac{1}{2\pi |\mathbf{R}_i|^{\frac{1}{2}}} \exp\left[-\frac{1}{2}(\hat{\mathbf{a}}_i^{2D} - \mathbf{a}^{2D})^T \mathbf{R}_i^{-1} (\hat{\mathbf{a}}_i^{2D} - \mathbf{a}^{2D})\right],
\end{aligned} \tag{47}$$

where $\hat{\mathbf{a}}_i^{2D}$ and $\mathbf{a}^{2D}$ denotes the 2×1 vectors of $\hat{\mathbf{a}}_i$'s and $\mathbf{a}$'s first two elements, respectively; $\mathbf{R}_i$ is expressed as

$$\mathbf{R}_i = \frac{3\sigma_n^2}{Mp_{\text{max},i}^2} \begin{bmatrix} \frac{h^2(r_{i,1}^2 + r_{i,3}^2)}{r_{i,3}^4} & \frac{h^2 r_{i,1} r_{i,2}}{r_{i,3}^4} \\ \frac{h^2 r_{i,1} r_{i,2}}{r_{i,3}^4} & \frac{h^2(r_{i,2}^2 + r_{i,3}^2)}{r_{i,3}^4} \end{bmatrix}. \tag{48}$$

Proof. With (7) and (19), we calculate the expectation of $\hat{\mathbf{a}}_i - \mathbf{a}$ as

$$E(\hat{\mathbf{a}}_i - \mathbf{a}) = \frac{3}{Mp_{\text{max},i}} \mathbf{A}_i \mathbf{V}_{\text{PD}}^T E(\mathbf{n}_i) = \mathbf{0}_3. \tag{49}$$

According to (19), the covariance matrix of $\hat{\mathbf{a}}_i - \mathbf{a}$ is expressed as

$$E[(\hat{\mathbf{a}}_i - \mathbf{a})(\hat{\mathbf{a}}_i - \mathbf{a})^T] = \mathbf{A}_i E(\mathbf{n}_i \mathbf{n}_i^T) \mathbf{A}_i^T. \tag{50}$$

Substituting (8) into (50), we obtain

$$E[(\hat{\mathbf{a}}_i - \mathbf{a})(\hat{\mathbf{a}}_i - \mathbf{a})^T] = \frac{3\sigma_n^2}{Mp_{\text{max},i}^2} \mathbf{A}_i \mathbf{A}_i^T. \tag{51}$$

Applying $\mathbf{A}_i$ in (18) to (51), we have

$$\begin{aligned}
E[(\hat{\mathbf{a}}_i - \mathbf{a})(\hat{\mathbf{a}}_i - \mathbf{a})^T] \\
= \frac{3\sigma_n^2}{Mp_{\text{max},i}^2} \begin{bmatrix} \frac{h^2(r_{i,1}^2 + r_{i,3}^2)}{r_{i,3}^4} & \frac{h^2 r_{i,1} r_{i,2}}{r_{i,3}^4} & 0 \\ \frac{h^2 r_{i,1} r_{i,2}}{r_{i,3}^4} & \frac{h^2(r_{i,2}^2 + r_{i,3}^2)}{r_{i,3}^4} & 0 \\ 0 & 0 & 0 \end{bmatrix}.
\end{aligned} \tag{52}$$

The third row and the third column of the matrix in (52) are all zero items, which is because $\hat{\mathbf{a}}_i$ is a 2D estimation, thus the z -axis item of $\hat{\mathbf{a}}_i$ contains no error. These zero items in (52) makes this matrix singular, and thus the inverse matrix of (52) does not exist. Therefore, for the convenience of the following derivations, we define the 2×1 vectors of $\hat{\mathbf{a}}_i$'s and $\mathbf{a}$'s first two elements as $\hat{\mathbf{a}}_i^{2D}$ and $\mathbf{a}^{2D}$, respectively. Based on (49) and (52), we have

$$E(\hat{\mathbf{a}}_i^{2D} - \mathbf{a}^{2D}) = \mathbf{0}_2, \tag{53}$$

and

$$\begin{aligned}
\mathbf{R}_i &:= E[(\hat{\mathbf{a}}_i^{2D} - \mathbf{a}^{2D})(\hat{\mathbf{a}}_i^{2D} - \mathbf{a}^{2D})^T] \\
&= \frac{3\sigma_n^2}{Mp_{\text{max},i}^2} \begin{bmatrix} \frac{h^2(r_{i,1}^2 + r_{i,3}^2)}{r_{i,3}^4} & \frac{h^2 r_{i,1} r_{i,2}}{r_{i,3}^4} \\ \frac{h^2 r_{i,1} r_{i,2}}{r_{i,3}^4} & \frac{h^2(r_{i,2}^2 + r_{i,3}^2)}{r_{i,3}^4} \end{bmatrix}
\end{aligned} \tag{54}$$

which is a non-singular matrix. Based on (53) and (54), $\hat{\mathbf{a}}_i^{2D}$ follows a 2D Gaussian distribution as

$$\begin{aligned}
\Pr(\hat{\mathbf{a}}_i^{2D} | \mathbf{a}^{2D}) \\
= \frac{1}{2\pi |\mathbf{R}_i|^{\frac{1}{2}}} \exp\left[-\frac{1}{2}(\hat{\mathbf{a}}_i^{2D} - \mathbf{a}^{2D})^T \mathbf{R}_i^{-1} (\hat{\mathbf{a}}_i^{2D} - \mathbf{a}^{2D})\right],
\end{aligned} \tag{55}$$

which is the probability of $\hat{\mathbf{a}}_i^{2D}$ when given $\mathbf{a}^{2D}$. $\square$

Based on **Theorem V.1** and maximum likelihood criterion, we can formulate an optimization problem to derive the MLC method.

Theorem V.2 (MLC localization result). *The localization result obtained with MLC is expressed as*

$$\hat{\mathbf{a}}_{\text{mlc}} = \left(\sum_{i=1}^N \hat{\mathbf{G}}_i^{-1} \right)^{-1} \left(\sum_{i=1}^N \hat{\mathbf{G}}_i^{-1} \hat{\mathbf{a}}_i^{2\text{D}} \right), \quad (56)$$

where

$$\hat{\mathbf{G}}_i = \begin{bmatrix} \frac{\hat{r}_{i,1}^2 + \hat{r}_{i,3}^2}{\hat{r}_{i,3}^{2m+8}} & \frac{\hat{r}_{i,1} \hat{r}_{i,2}}{\hat{r}_{i,3}^{2m+8}} \\ \frac{\hat{r}_{i,1} \hat{r}_{i,2}}{\hat{r}_{i,3}^{2m+8}} & \frac{\hat{r}_{i,2}^2 + \hat{r}_{i,3}^2}{\hat{r}_{i,3}^{2m+8}} \end{bmatrix}. \quad (57)$$

Proof. Based on **Theorem V.1**, a likelihood function is formulated as

$$f_L = \prod_{i=1}^N \Pr(\hat{\mathbf{a}}_i^{2\text{D}} | \mathbf{a}^{2\text{D}}). \quad (58)$$

To derive the estimation of $\mathbf{a}^{2\text{D}}$, we formulate the optimization problem as follow

$$\begin{aligned} \hat{\mathbf{a}}^{2\text{D}} &= \arg \max_{\mathbf{a}^{2\text{D}}} \prod_{i=1}^N \Pr(\hat{\mathbf{a}}_i^{2\text{D}} | \mathbf{a}^{2\text{D}}) \\ &= \arg \max_{\mathbf{a}^{2\text{D}}} \sum_{i=1}^N \ln[\Pr(\hat{\mathbf{a}}_i^{2\text{D}} | \mathbf{a}^{2\text{D}})] \\ &= \arg \min_{\mathbf{a}^{2\text{D}}} \sum_{i=1}^N (\hat{\mathbf{a}}_i^{2\text{D}} - \mathbf{a}^{2\text{D}})^T \mathbf{R}_i^{-1} (\hat{\mathbf{a}}_i^{2\text{D}} - \mathbf{a}^{2\text{D}}). \end{aligned} \quad (59)$$

This optimization problem can be solved by setting

$$\begin{aligned} &\frac{\partial \sum_{i=1}^N (\hat{\mathbf{a}}_i^{2\text{D}} - \mathbf{a}^{2\text{D}})^T \mathbf{R}_i^{-1} (\hat{\mathbf{a}}_i^{2\text{D}} - \mathbf{a}^{2\text{D}})}{\partial \mathbf{a}^{2\text{D}}} \\ &= \sum_{i=1}^N 2\mathbf{R}_i^{-1} (\hat{\mathbf{a}}_i^{2\text{D}} - \mathbf{a}^{2\text{D}}) = \mathbf{0}_2. \end{aligned} \quad (60)$$

Therefore, $\mathbf{a}^{2\text{D}}$ is estimated as

$$\hat{\mathbf{a}}^{2\text{D}} = \left(\sum_{i=1}^N \mathbf{R}_i^{-1} \right)^{-1} \left(\sum_{i=1}^N \mathbf{R}_i^{-1} \hat{\mathbf{a}}_i^{2\text{D}} \right). \quad (61)$$

However, since $\mathbf{R}_i$ in (54) is not available, we should apply $\hat{\mathbf{r}}_i = [\hat{r}_{i,1}, \hat{r}_{i,2}, \hat{r}_{i,3}]^T$ to (54) and obtain the estimation of $\mathbf{R}_i$ as

$$\hat{\mathbf{R}}_i = \frac{3\sigma_n^2}{Mp_{\max,i}^2} \begin{bmatrix} \frac{h^2(\hat{r}_{i,1}^2 + \hat{r}_{i,3}^2)}{\hat{r}_{i,3}^4} & \frac{h^2 \hat{r}_{i,1} \hat{r}_{i,2}}{\hat{r}_{i,3}^4} \\ \frac{h^2 \hat{r}_{i,1} \hat{r}_{i,2}}{\hat{r}_{i,3}^4} & \frac{h^2(\hat{r}_{i,2}^2 + \hat{r}_{i,3}^2)}{\hat{r}_{i,3}^4} \end{bmatrix} \quad (62)$$

where $\hat{p}_{\max,i}$ is the estimation of $p_{\max,i}$. Since we assume that the LEDs have the same normal vector $[0, 0, -1]^T$, $\hat{p}_{\max,i}$ is expressed as

$$\hat{p}_{\max,i} = P_t \frac{(m+1)\alpha}{2\pi \hat{d}_i^2} \left(\frac{h}{\hat{d}_i} \right)^m = P_t \frac{(m+1)\alpha h^m}{2\pi \hat{d}_i^{m+2}} \quad (63)$$

where $\hat{d}_i = \|\mathbf{t}_i - \hat{\mathbf{a}}\|$ is the estimation of d_i . Since $\hat{\mathbf{r}}_i = (\mathbf{t}_i - \hat{\mathbf{a}})/\|\mathbf{t}_i - \hat{\mathbf{a}}\|$, we have

$$\hat{r}_{i,3} = \frac{h}{\hat{d}_i}. \quad (64)$$

Applying (64) to (63), we calculate $\hat{p}_{\max,i}$ as

$$\hat{p}_{\max,i} = P_t \frac{(m+1)\alpha \hat{r}_{i,3}^{m+2}}{2\pi h^2}. \quad (65)$$

Substituting (65) into (62), we obtain

$$\hat{\mathbf{R}}_i = \frac{12\pi^2 \sigma_n^2 h^6}{MP_t^2 (m+1)^2 \alpha^2} \underbrace{\begin{bmatrix} \frac{\hat{r}_{i,1}^2 + \hat{r}_{i,3}^2}{\hat{r}_{i,3}^{2m+8}} & \frac{\hat{r}_{i,1} \hat{r}_{i,2}}{\hat{r}_{i,3}^{2m+8}} \\ \frac{\hat{r}_{i,1} \hat{r}_{i,2}}{\hat{r}_{i,3}^{2m+8}} & \frac{\hat{r}_{i,2}^2 + \hat{r}_{i,3}^2}{\hat{r}_{i,3}^{2m+8}} \end{bmatrix}}_{\hat{\mathbf{G}}_i} \quad (66)$$

where $12\pi^2 \sigma_n^2 h^6 / (MP_t^2 (m+1)^2 \alpha^2)$ is a constant. Replacing $\mathbf{R}_i$ in (61) with $\hat{\mathbf{R}}_i$ in (66), we have

$$\begin{aligned} \hat{\mathbf{a}}_{\text{mlc}} &= \left(\sum_{i=1}^N \hat{\mathbf{R}}_i^{-1} \right)^{-1} \left(\sum_{i=1}^N \hat{\mathbf{R}}_i^{-1} \hat{\mathbf{a}}_i^{2\text{D}} \right) \\ &= \left(\sum_{i=1}^N \hat{\mathbf{G}}_i^{-1} \right)^{-1} \left(\sum_{i=1}^N \hat{\mathbf{G}}_i^{-1} \hat{\mathbf{a}}_i^{2\text{D}} \right) \end{aligned} \quad (67)$$

which is the localization result with the MLC. $\square$

This indicates that the MLC method can work without the exact values of the system parameters, such as σ_n and P_t . Since $\hat{\mathbf{G}}_i$ is a 2×2 matrix, its inverse matrix can be easily calculated, and thus the MLC method proposed in (67) has a low complexity, which contains $2(N-1)$ matrix additions, $N+1$ matrix multiplication, and $N+1$ inversion operations of 2×2 matrices. As shown in (11), $\hat{\mathbf{a}}_i = \mathbf{a}$ when $\hat{\mathbf{r}}_i = \mathbf{r}_i$, which indicates $\hat{\mathbf{a}}_i^{2\text{D}} = \mathbf{a}^{2\text{D}}$ when $\hat{\mathbf{r}}_i = \mathbf{r}_i$. Applying $\hat{\mathbf{a}}_i^{2\text{D}} = \mathbf{a}^{2\text{D}}$ to (67), we have

$$\hat{\mathbf{a}}_{\text{mlc}} = \left(\sum_{i=1}^N \hat{\mathbf{G}}_i^{-1} \right)^{-1} \left(\sum_{i=1}^N \hat{\mathbf{G}}_i^{-1} \right) \mathbf{a}^{2\text{D}} = \mathbf{a}^{2\text{D}}, \quad (68)$$

which means that $\hat{\mathbf{a}}_{\text{mlc}} = \mathbf{a}^{2\text{D}}$ when $\hat{\mathbf{r}}_1 = \mathbf{r}_1, \hat{\mathbf{r}}_2 = \mathbf{r}_2, \dots, \hat{\mathbf{r}}_N = \mathbf{r}_N$. The proposed MLC method in (67) makes full use of the statistics of the AOA samples to achieve the highest possible accuracy under the maximum likelihood criterion.

C. Positioning Error With MLC

Theorem V.3 (MLC localization error). *The localization error of the MLC method is expressed as*

$$e_{\text{mlc}} = \sqrt{\sum_{i=1}^N \frac{3\sigma_n^2}{Mp_{\max,i}^2} \text{tr}(\mathbf{H}_i \mathbf{H}_i^T)}, \quad (69)$$

where $\mathbf{H}_i$ is defined in (79).

Proof. To derive the estimation error of $\hat{\mathbf{a}}_{\text{mlc}}$, we need to apply Taylor series expansion to (67) at $\hat{\mathbf{r}}_1 = \mathbf{r}_1, \dots, \hat{\mathbf{r}}_N = \mathbf{r}_N$ as

$$\begin{aligned} \hat{\mathbf{a}}_{\text{mlc}} &= \mathbf{a}^{2\text{D}} + \sum_{i=1}^N \frac{\partial \hat{\mathbf{a}}_{\text{mlc}}}{\partial \hat{\mathbf{r}}_i} \bigg|_{\hat{\mathbf{r}}_1=\mathbf{r}_1, \dots, \hat{\mathbf{r}}_N=\mathbf{r}_N} (\hat{\mathbf{r}}_i - \mathbf{r}_i) + o(\hat{\mathbf{r}}_i - \mathbf{r}_i) \\ &\approx \mathbf{a}^{2\text{D}} + \sum_{i=1}^N \frac{\partial \hat{\mathbf{a}}_{\text{mlc}}}{\partial \hat{\mathbf{r}}_i} \bigg|_{\hat{\mathbf{r}}_1=\mathbf{r}_1, \dots, \hat{\mathbf{r}}_N=\mathbf{r}_N} \mathbf{n}_i^r \end{aligned} \quad (70)$$

where $o(\hat{\mathbf{r}}_i - \mathbf{r}_i)$ denotes a vector function whose elements are the higher order small quantities of $\hat{\mathbf{r}}_i - \mathbf{r}_i$. If $\hat{\mathbf{a}}_{\text{mlc}} = [\hat{a}_{\text{mlc},1}, \hat{a}_{\text{mlc},2}]^T$, $\partial\hat{\mathbf{a}}_{\text{mlc}}/\partial\hat{\mathbf{r}}_i$ is expressed as

$$\begin{aligned} \frac{\partial\hat{\mathbf{a}}_{\text{mlc}}}{\partial\hat{\mathbf{r}}_i} &= \begin{bmatrix} \frac{\partial\hat{\mathbf{a}}_{\text{mlc}}}{\partial\hat{r}_{i,1}}, \frac{\partial\hat{\mathbf{a}}_{\text{mlc}}}{\partial\hat{r}_{i,2}}, \frac{\partial\hat{\mathbf{a}}_{\text{mlc}}}{\partial\hat{r}_{i,3}} \end{bmatrix} \\ &= \begin{bmatrix} \frac{\partial\hat{a}_{\text{mlc},1}}{\partial\hat{r}_{i,1}} & \frac{\partial\hat{a}_{\text{mlc},1}}{\partial\hat{r}_{i,2}} & \frac{\partial\hat{a}_{\text{mlc},1}}{\partial\hat{r}_{i,3}} \\ \frac{\partial\hat{a}_{\text{mlc},2}}{\partial\hat{r}_{i,1}} & \frac{\partial\hat{a}_{\text{mlc},2}}{\partial\hat{r}_{i,2}} & \frac{\partial\hat{a}_{\text{mlc},2}}{\partial\hat{r}_{i,3}} \end{bmatrix}. \end{aligned} \quad (71)$$

According to (67), $\partial\hat{\mathbf{a}}_{\text{mlc}}/\partial\hat{r}_{i,j}$ is calculated as

$$\begin{aligned} \frac{\partial\hat{\mathbf{a}}_{\text{mlc}}}{\partial\hat{r}_{i,j}} &= \frac{\partial(\sum_{i=1}^N \hat{\mathbf{G}}_i^{-1})^{-1}}{\partial\hat{r}_{i,j}} \left(\sum_{i=1}^N \hat{\mathbf{G}}_i^{-1} \hat{\mathbf{a}}_i^{2D} \right) \\ &\quad + \left(\sum_{i=1}^N \hat{\mathbf{G}}_i^{-1} \right)^{-1} \frac{\partial \sum_{i=1}^N \hat{\mathbf{G}}_i^{-1} \hat{\mathbf{a}}_i^{2D}}{\partial\hat{r}_{i,j}} \\ &= \frac{\partial(\sum_{i=1}^N \hat{\mathbf{G}}_i^{-1})^{-1}}{\partial\hat{r}_{i,j}} \left(\sum_{i=1}^N \hat{\mathbf{G}}_i^{-1} \hat{\mathbf{a}}_i^{2D} \right) \\ &\quad + \left(\sum_{i=1}^N \hat{\mathbf{G}}_i^{-1} \right)^{-1} \left(\frac{\partial\hat{\mathbf{G}}_i^{-1}}{\partial\hat{r}_{i,j}} \hat{\mathbf{a}}_i^{2D} + \hat{\mathbf{G}}_i^{-1} \frac{\partial\hat{\mathbf{a}}_i^{2D}}{\partial\hat{r}_{i,j}} \right). \end{aligned} \quad (72)$$

Based on [30, eq. (59)], eq. (72) is calculated as

$$\begin{aligned} \frac{\partial\hat{\mathbf{a}}_{\text{mlc}}}{\partial\hat{r}_{i,j}} &= - \left(\sum_{i=1}^N \hat{\mathbf{G}}_i^{-1} \right)^{-1} \frac{\partial \sum_{i=1}^N \hat{\mathbf{G}}_i^{-1}}{\partial\hat{r}_{i,j}} \left(\sum_{i=1}^N \hat{\mathbf{G}}_i^{-1} \right)^{-1} \\ &\quad \times \left(\sum_{i=1}^N \hat{\mathbf{G}}_i^{-1} \hat{\mathbf{a}}_i^{2D} \right) \\ &\quad + \left(\sum_{i=1}^N \hat{\mathbf{G}}_i^{-1} \right)^{-1} \left(\frac{\partial\hat{\mathbf{G}}_i^{-1}}{\partial\hat{r}_{i,j}} \hat{\mathbf{a}}_i^{2D} + \hat{\mathbf{G}}_i^{-1} \frac{\partial\hat{\mathbf{a}}_i^{2D}}{\partial\hat{r}_{i,j}} \right) \\ &= \left(\sum_{i=1}^N \hat{\mathbf{G}}_i^{-1} \right)^{-1} \left(\hat{\mathbf{G}}_i^{-1} \frac{\partial\hat{\mathbf{G}}_i}{\partial\hat{r}_{i,j}} \hat{\mathbf{G}}_i^{-1} \right) \left(\sum_{i=1}^N \hat{\mathbf{G}}_i^{-1} \right)^{-1} \\ &\quad \times \left(\sum_{i=1}^N \hat{\mathbf{G}}_i^{-1} \hat{\mathbf{a}}_i^{2D} \right) \\ &\quad + \left(\sum_{i=1}^N \hat{\mathbf{G}}_i^{-1} \right)^{-1} \left(-\hat{\mathbf{G}}_i^{-1} \frac{\partial\hat{\mathbf{G}}_i}{\partial\hat{r}_{i,j}} \hat{\mathbf{G}}_i^{-1} \hat{\mathbf{a}}_i^{2D} + \hat{\mathbf{G}}_i^{-1} \frac{\partial\hat{\mathbf{a}}_i^{2D}}{\partial\hat{r}_{i,j}} \right). \end{aligned} \quad (73)$$

To obtain (73), we need to calculate $\partial\hat{\mathbf{G}}_i/\partial\hat{r}_{i,j}$ and $\partial\hat{\mathbf{a}}_i^{2D}/\partial\hat{r}_{i,j}$. According to (66), we have

$$\frac{\partial\hat{\mathbf{G}}_i}{\partial\hat{r}_{i,1}} = \begin{bmatrix} \frac{2\hat{r}_{i,1}}{\hat{r}_{i,3}^{2m+8}} & \frac{\hat{r}_{i,2}}{\hat{r}_{i,3}^{2m+8}} \\ \frac{\hat{r}_{i,2}}{\hat{r}_{i,3}^{2m+8}} & 0 \end{bmatrix}, \quad (74)$$

$$\frac{\partial\hat{\mathbf{G}}_i}{\partial\hat{r}_{i,2}} = \begin{bmatrix} 0 & \frac{\hat{r}_{i,1}}{\hat{r}_{i,3}^{2m+8}} \\ \frac{\hat{r}_{i,1}}{\hat{r}_{i,3}^{2m+8}} & \frac{2\hat{r}_{i,2}}{\hat{r}_{i,3}^{2m+8}} \end{bmatrix}, \quad (75)$$

$$\begin{aligned} \frac{\partial\hat{\mathbf{G}}_i}{\partial\hat{r}_{i,3}} &= \\ &\begin{bmatrix} -\frac{(2m+8)\hat{r}_{i,1}^2 + (2m+6)\hat{r}_{i,3}^2}{\hat{r}_{i,3}^{2m+9}} & -\frac{(2m+8)\hat{r}_{i,1}\hat{r}_{i,2}}{\hat{r}_{i,3}^{2m+9}} \\ -\frac{(2m+8)\hat{r}_{i,1}\hat{r}_{i,2}}{\hat{r}_{i,3}^{2m+9}} & -\frac{(2m+8)\hat{r}_{i,2}^2 + (2m+6)\hat{r}_{i,3}^2}{\hat{r}_{i,3}^{2m+9}} \end{bmatrix}. \end{aligned} \quad (76)$$

According to (10), $\partial\hat{\mathbf{a}}_i^{2D}/\partial\hat{r}_{i,j}$ is calculated as

$$\begin{aligned} \frac{\partial\hat{\mathbf{a}}_i^{2D}}{\partial\hat{r}_{i,1}} &= \begin{bmatrix} -\frac{h}{\hat{r}_{i,3}}, 0 \end{bmatrix}^T, \\ \frac{\partial\hat{\mathbf{a}}_i^{2D}}{\partial\hat{r}_{i,2}} &= \begin{bmatrix} 0, -\frac{h}{\hat{r}_{i,3}} \end{bmatrix}^T, \\ \frac{\partial\hat{\mathbf{a}}_i^{2D}}{\partial\hat{r}_{i,3}} &= \begin{bmatrix} \frac{h\hat{r}_{i,1}}{\hat{r}_{i,3}^2}, \frac{h\hat{r}_{i,2}}{\hat{r}_{i,3}^2} \end{bmatrix}^T. \end{aligned} \quad (77)$$

We substitute (74)-(77) into (73) to obtain $\partial\hat{\mathbf{a}}_{\text{mlc}}/\partial\hat{r}_{i,j}$ s, and then substitute them into (71) to calculate $\partial\hat{\mathbf{a}}_{\text{mlc}}/\partial\hat{\mathbf{r}}_i$. With $\partial\hat{\mathbf{a}}_{\text{mlc}}/\partial\hat{\mathbf{r}}_i$, we can finally calculate (70) and express it as

$$\hat{\mathbf{a}}_{\text{mlc}} \approx \mathbf{a}^{2D} + \sum_{i=1}^N \mathbf{H}_i \mathbf{n}_i^r \quad (78)$$

where

$$\mathbf{H}_i := \left. \frac{\partial\hat{\mathbf{a}}_{\text{mlc}}}{\partial\hat{\mathbf{r}}_i} \right|_{\hat{\mathbf{r}}_1=\mathbf{r}_1, \dots, \hat{\mathbf{r}}_N=\mathbf{r}_N}. \quad (79)$$

Subtracting $\mathbf{a}^{2D}$ from both sides of (78), we have

$$\hat{\mathbf{a}}_{\text{mlc}} - \mathbf{a}^{2D} \approx \sum_{i=1}^N \mathbf{H}_i \mathbf{n}_i^r, \quad (80)$$

which denotes the estimation error vector of $\hat{\mathbf{a}}_{\text{mlc}}$. According to (7), the MSE of $\hat{\mathbf{a}}_{\text{mlc}}$ is calculated as

$$\begin{aligned} e_{\text{mlc}}^2 &= \mathbb{E}[(\hat{\mathbf{a}}_{\text{mlc}} - \mathbf{a}^{2D})^T (\hat{\mathbf{a}}_{\text{mlc}} - \mathbf{a}^{2D})] \\ &= \mathbb{E} \left[\left(\sum_{i=1}^N \frac{3\mathbf{n}_i^T \mathbf{V}_{\text{PD}} \mathbf{H}_i^T}{M p_{\text{max},i}} \right) \left(\sum_{i=1}^N \frac{3\mathbf{H}_i \mathbf{V}_{\text{PD}}^T \mathbf{n}_i}{M p_{\text{max},i}} \right) \right] \\ &= \sum_{i=1}^N \frac{9}{M^2 p_{\text{max},i}^2} \mathbb{E}(\mathbf{n}_i^T \mathbf{V}_{\text{PD}} \mathbf{H}_i^T \mathbf{H}_i \mathbf{V}_{\text{PD}}^T \mathbf{n}_i) \\ &= \sum_{i=1}^N \frac{3\sigma_n^2}{M p_{\text{max},i}^2} \text{tr}(\mathbf{H}_i \mathbf{H}_i^T) \end{aligned} \quad (81)$$

where e_{mlc} quantifies the positioning error of the MLC, which is calculated as

$$e_{\text{mlc}} = \sqrt{\sum_{i=1}^N \frac{3\sigma_n^2}{M p_{\text{max},i}^2} \text{tr}(\mathbf{H}_i \mathbf{H}_i^T)}. \quad (82)$$

□

Equation (69) relates the MLC positioning error with the system parameters, and can help circumvent the time-consuming Monte Carlo simulations.

In this paper, we exploit low-cost PDs and propose low-complexity methods for positioning, and focus on the scenario of strong line-of-sight (LoS) signals and weak non-LoS signals. For the scenario of strong non-LoS signals, we had designed a special mode-mixed PD array to extract the LoS information and suppress the interference of the non-LoS signals [32].

Last but not the least, the localization methods proposed in (44) and (56) are all in summation forms, and the LEDs blocked by obstacles can be omitted by ID detection [34]. It means that the proposed combining methods can automatically adjust the summation terms according to the channel state

information, which guarantees the robustness of the VLP system.

VI. SIMULATION RESULTS

The LED's transmission power is $P_t = 20$ W, and its Lambertian order is set as $m = 1$ to ensure a sufficient field-of-view. Since the LED faces the ground, its normal vector is $[0, 0, -1]^T$. The height of the indoor space is set as $h = 4$ m, in other words, the z -axis coordinate of the LED is 4 m. The AOA estimator contains $M = 4$ PDs, and each PD has an active area of $\alpha = 1$ cm². Last but not the least, the standard deviation of the noise sample is set as $\sigma_n = 5 \times 10^{-7}$ W to test the performances of the proposed localization methods.

A. Localization Results of the Proposed Methods

As shown in Fig. 2, we exploit the positioning methods of the single-LED in (9), LED-select in (26), EGC in (44) and MLC in (67) to localize an AOA estimator, which is set at 16 different positions of an 8×8 m² area for testing. In Fig. 2(a), the localization results obtained with a single LED are demonstrated, where the LED is set at the room center $[4 \text{ m}, 4 \text{ m}, 4 \text{ m}]^T$. Near the LED, the localization results cluster tightly around the exact positions of the AOA estimator, which have small localization errors. At the corners of the indoor space, the localization results deviate greatly from the exact positions of the AOA estimator, which exhibit big localization errors. These results agree with the analytical derivation in (14) that the single-LED localization suffers from a bigger positioning error when the AOA estimator is farther from the LED. In Figs. 2(b)-2(d), four LEDs are set at the centers of four 4×4 m² subareas. As shown in Fig. 2(b), the localization results obtained with the LED selection are closely distributed around the exact positions of the AOA estimator, which verify that the proposed LED selection method can effectively improve the positioning accuracy. In Fig. 2(c), the localization results of the EGC are demonstrated. Though the localization errors are small around the room center, the positioning performance at the room corners is even worse than that of Fig. 2(a). This can be explained with the analysis at the end of Sec. V-A, where few weak LED signals can degrade the localization accuracy of the EGC. Taking the positioning at $[7 \text{ m}, 1 \text{ m}, 0 \text{ m}]^T$ as an example, we find that though the LED at $[6 \text{ m}, 2 \text{ m}, 4 \text{ m}]^T$ is close to the AOA estimator, the other three LEDs are far away from it. Therefore, the localization errors of these three LEDs would significantly increase e_{egc} in (46), resulting a big localization error. In Fig. 2(d), the localization results obtained with the MLC are shown, which are similar to those in Fig. 2(b) and have small localization errors. In summary, we have following insights about these localization methods:

- When an LED is set at the room center, the single-LED method has small errors around the room center and big errors at the room corners.
- The LED-select and MLC methods can significantly improve the positioning accuracy.
- The EGC method may reduce the localization accuracy when the indoor area is sufficiently large and the LEDs are sparsely deployed.

B. Positioning Errors of the Proposed Methods

In this section, we would like to verify the theoretical positioning error expressions in (14), (36), (46) and (69) within a $4 \times 4 \times 4$ m³ indoor space. To quantitatively characterize the localization performance of the proposed methods, we define the average value and the mean square deviation of their positioning errors over the entire indoor space as

$$e_{\text{ave}} = \int_0^4 \int_0^4 \frac{e_{\text{pm}}(x, y)}{4 \times 4} dx dy = \int_0^4 \int_0^4 \frac{e_{\text{pm}}(x, y)}{16} dx dy \quad (83)$$

and

$$e_{\text{var}} = \int_0^4 \int_0^4 \frac{[e_{\text{pm}}(x, y) - e_{\text{ave}}]^2}{16} dx dy \quad (84)$$

where $e_{\text{pm}}(x, y)$ denotes the positioning error of a proposed method when the AOA estimator is at $[x, y, 0 \text{ m}]^T$; e_{ave} and e_{var} denote the average value and mean square deviation of the positioning errors over the 4×4 m² indoor area, respectively. A low e_{ave} value guarantees a low overall localization error in the whole area, and a low e_{var} value guarantees the uniformity of the error level at different locations.

In Figs. 3-6, we plot the positioning errors of the single-LED, LED-select, EGC and MLC methods, which are obtained with the theoretical derivations and Monte Carlo simulations. The ratios of the positioning errors obtained with these two ways are plotted in each subfigure (c), which fluctuate slightly around 1. This indicates that the theoretical positioning error results derived in (14), (36), (46) and (69) agree with the Monte Carlo simulation results. In Fig. 3 and Figs. 5-6, the positioning error of the AOA estimator at $[x, y, 0 \text{ m}]^T$ has a similar trend, where the positioning error is minimized around the room center and maximized at the corners. A different trend is shown in Fig. 4, where the positioning error exhibits a wave-like shape containing two minimums. This is because two LED are exploited in Fig. 4, and thus the positioning error results tend to be the fusion of two Fig. 3-like figures with different minimum locations. Moreover, according to e_{ave} listed below each subfigure (a), we can rank the localization performance of these methods as $\text{MLC} > \text{EGC} > \text{LED-select} > \text{single-LED}$, where $>$ means "is superior to". This ranking of EGC is different from that of Fig. 2, which is due to the change of the indoor space size. In Fig. 5, the room size is shrunk to $4 \times 4 \times 4$ m³, which is more suitable for EGC to combine the single-LED positioning results to reduce the overall positioning error. Last but not the least, as listed below Fig. 6(a), $e_{\text{ave}} = 4.77$ cm, which achieves a centimeter-level accuracy and is much smaller than that of Fig. 3.

C. Relationship Between Positioning Error and LED Number

In this section, we aim to investigate how positioning error changes with the LED number N . For convenience, the LED number is set as $N = 1^2, 2^2, 3^2, \dots$, and the LEDs are placed at $x, y = 2/\sqrt{N}$ m, $6/\sqrt{N}$ m, $\dots$, $2(2\sqrt{N} - 1)/\sqrt{N}$ m in the ceiling of a $4 \times 4 \times 4$ m³ indoor environment. In other words, the 4×4 m² ceiling is divided into N same squares, and the LEDs are in the centers of these squares. Besides, the

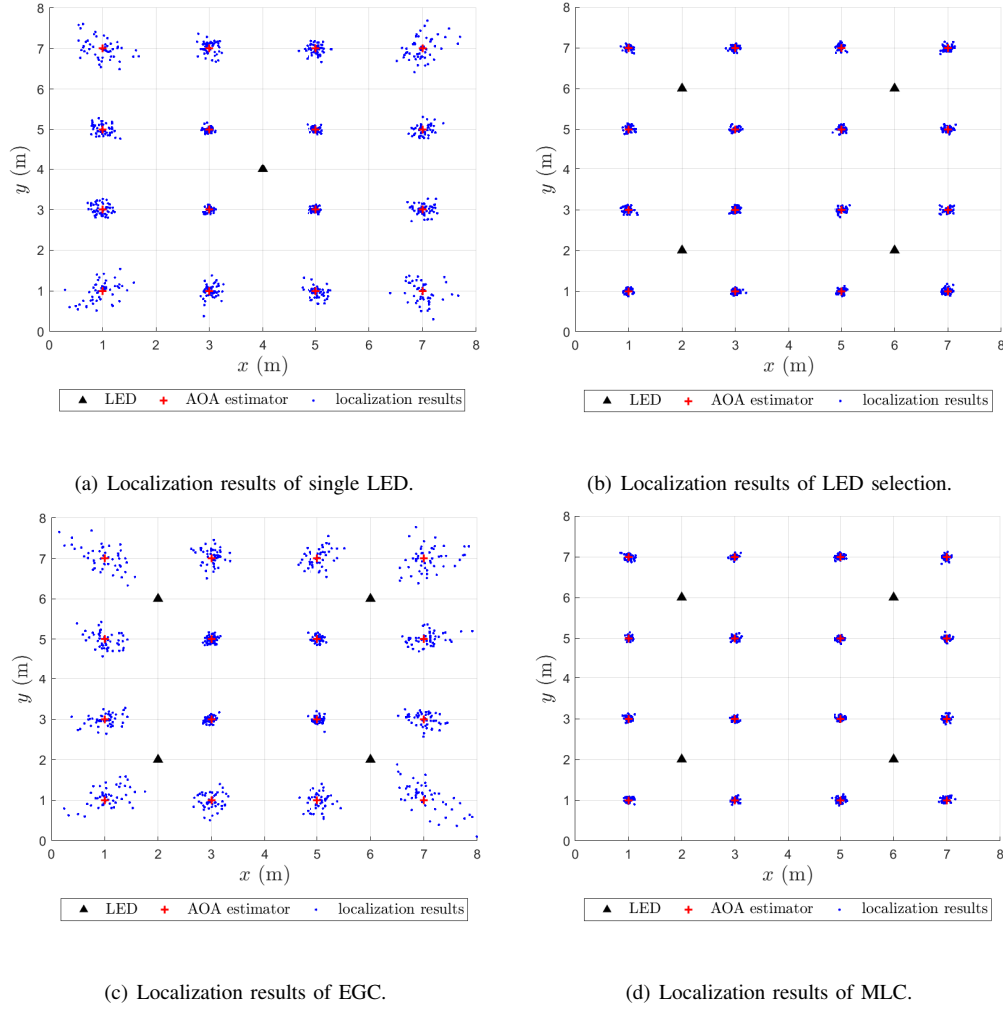


Fig. 2. Localization results obtained with the proposed methods. The AOA estimator is set at 16 different positions of an $8 \times 8 \text{ m}^2$ area for testing. In Fig. 2(a), the LED is set at the room center $[4 \text{ m}, 4 \text{ m}, 4 \text{ m}]^T$. In Figs. 2(b)-2(d), four LEDs are set at $[2 \text{ m}, 2 \text{ m}, 4 \text{ m}]^T$, $[2 \text{ m}, 6 \text{ m}, 4 \text{ m}]^T$, $[6 \text{ m}, 2 \text{ m}, 4 \text{ m}]^T$ and $[6 \text{ m}, 6 \text{ m}, 4 \text{ m}]^T$.

AOA estimators are placed at $x, y = 0 \text{ m}, 1 \text{ m}, 2 \text{ m}, 3 \text{ m}, 4 \text{ m}$ on the ground, which are $5 \times 5 = 25$ testing points in total. The average positioning error of these testing points are plotted as a function of $\sqrt{N}$ in Fig. 7 to demonstrate the overall localization performance of the proposed methods. Moreover, the results of the classic trilateration⁴ method and the angulation⁵ method are demonstrated in Fig. 7 for comparison. It should be noted that the single-LED method is exploited in the LED-select, EGC or MLC method when $N = 1$, while the trilateration method could not work with only one LED. As shown in Fig. 7, the curves have a similar trend of first decreasing and then gradually converging to a constant as $\sqrt{N}$ increases. This indicates that increasing the number of the LEDs can improve the positioning accuracy, but the improvement is negligible when N is sufficiently large. Moreover, when N is large, the average positioning errors of the EGC and MLC converge to smaller constants than that of

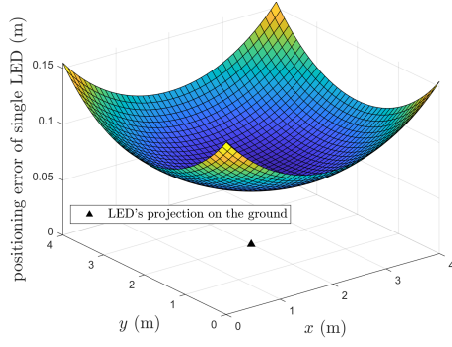
the LED-select. This indicates that the proposed combining methods tend to provide more accurate localization results over the $4 \times 4 \times 4 \text{ m}^3$ space when more LEDs are exploited. Besides, Fig. 7 verifies that (40) is the lower bound of the LED-select method's positioning error, which is relatively tight when $N \leq 25$. Last but not the least, Fig. 7 indicates that our proposed methods outperform the trilateration and the angulation in terms of the average positioning error over the $4 \times 4 \times 4 \text{ m}^3$ indoor environment.

D. Relationship Between Positioning Error and PDs' Tilted Angles

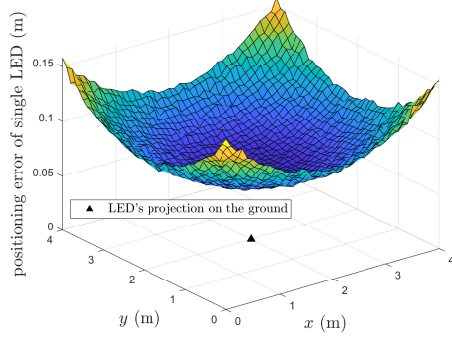
In this section, we would like to demonstrate how the positioning error of the single-LED localization changes with the tilted angles of the PDs. We assume that the four PDs are symmetrically placed around the z axis. Thus, the angles between the positive z axis and the PDs' normal vectors are exactly the same, which are denoted by ξ . Therefore, $\mathbf{V}_{PD}$ is

⁴The trilateration method is introduced in Sec. IV-A of [3], and its realization in this paper is demonstrated in Appendix B.

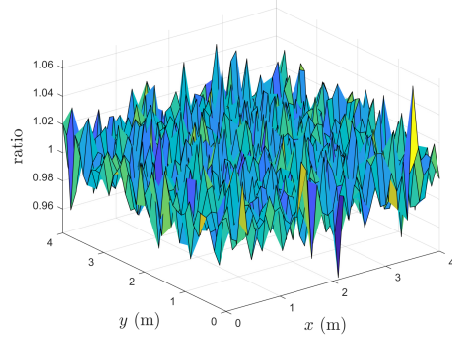
⁵The angulation method is introduced in Sec. V-B of [33], and its realization is demonstrated in Sec. III of [31].



(a) The positioning error of single LED derived in (14). $e_{ave} = 0.0898$ m. $e_{var} = 3.82 \times 10^{-4}$ m².



(b) The positioning error of single LED obtained with Monte Carlo simulations.



(c) The ratio of subfigure (b) and subfigure (a).

Fig. 3. The demonstration of single-LED method's positioning error obtained with (14) and Monte Carlo simulations. Their ratios are plotted in subfigure (c) to verify the theoretical result in (14). The LED is set at $[2 \text{ m}, 2 \text{ m}, 4 \text{ m}]^T$, and the AOA estimator is set at $[x, y, 0 \text{ m}]^T$.

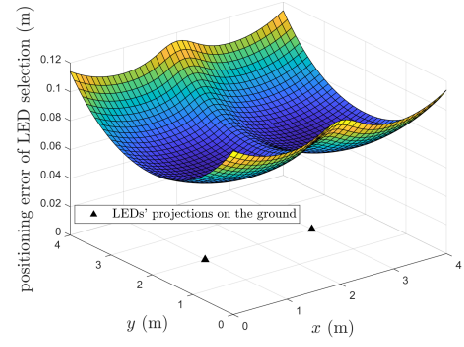
expressed as

$$\mathbf{V}_{PD} = \begin{bmatrix} \sin \xi & 0 & \cos \xi \\ 0 & \sin \xi & \cos \xi \\ -\sin \xi & 0 & \cos \xi \\ 0 & -\sin \xi & \cos \xi \end{bmatrix}. \quad (85)$$

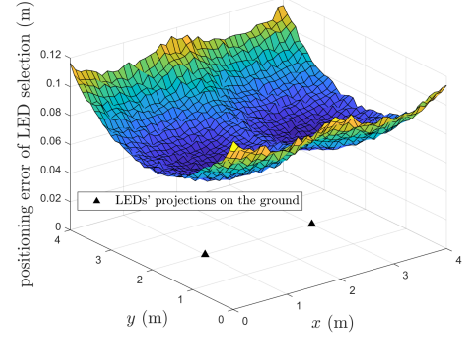
Since the optimal $\mathbf{V}_{PD}$ is

$$\mathbf{V}_{PD}^* = \sqrt{\frac{2}{3}} \begin{bmatrix} 1 & 0 & \frac{1}{\sqrt{2}} \\ 0 & 1 & \frac{1}{\sqrt{2}} \\ -1 & 0 & \frac{1}{\sqrt{2}} \\ 0 & -1 & \frac{1}{\sqrt{2}} \end{bmatrix}, \quad (86)$$

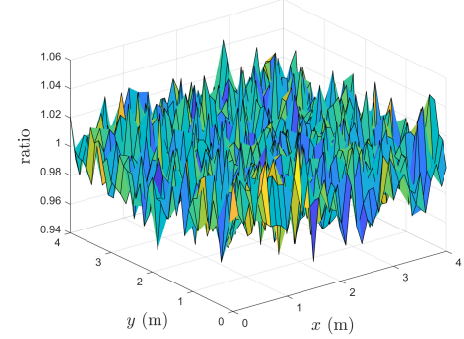
it is equivalent to the case of $\xi = 54.73^\circ$. Then, the indoor space is set as $4 \times 4 \times 4$ m³. The LED is set at $[2 \text{ m}, 2 \text{ m}, 4 \text{ m}]^T$, while the AOA estimator is set at



(a) The positioning error of LED selection derived in (36). $e_{ave} = 0.0787$ m. $e_{var} = 1.78 \times 10^{-4}$ m².



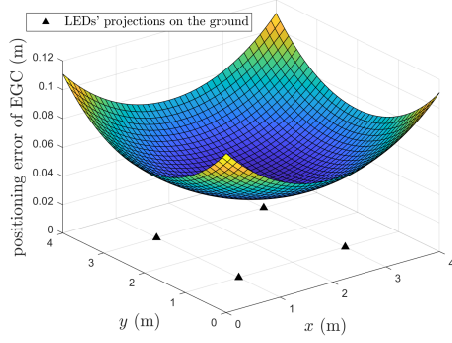
(b) The positioning error of LED selection obtained with Monte Carlo simulations.



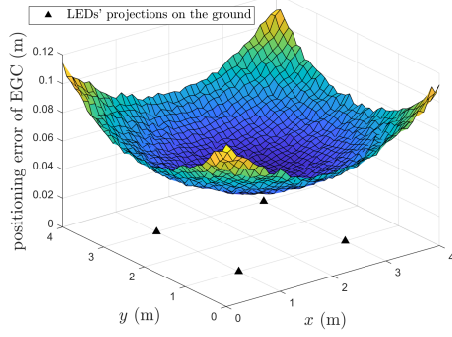
(c) The ratio of subfigure (b) and subfigure (a).

Fig. 4. The demonstration of LED-select method's positioning error obtained with (36) and Monte Carlo simulations. Their ratios are plotted in subfigure (c) to verify the theoretical result in (36). Two LEDs are set at $[1 \text{ m}, 2 \text{ m}, 4 \text{ m}]^T$ and $[3 \text{ m}, 2 \text{ m}, 4 \text{ m}]^T$. The AOA estimator is set at $[x, y, 0 \text{ m}]^T$.

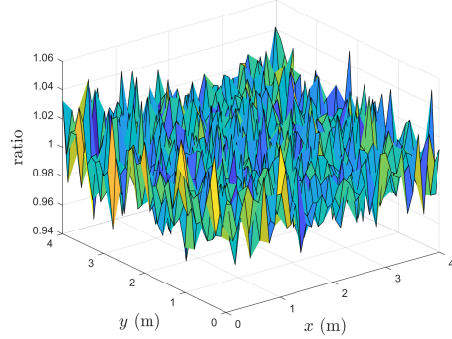
$[0 \text{ m}, 0 \text{ m}, 0 \text{ m}]^T$, $[1 \text{ m}, 1 \text{ m}, 0 \text{ m}]^T$ or $[2 \text{ m}, 2 \text{ m}, 0 \text{ m}]^T$. Under the aforementioned parameters, Fig. 8 demonstrates the positioning error with the single-LED localization when ξ changes. As shown in this figure, the curves have peaks around $\xi = 0^\circ$ and $\xi = 90^\circ$. This is because when $\xi = 0^\circ$, all the PDs face to the ceiling, and they can not provide enough diversities for the positioning; when $\xi = 90^\circ$, the PDs' normal vectors are all parallel to the ground, which makes received signals too weak to localize the AOA estimator. In other words, $\xi = 0^\circ$ or $\xi = 90^\circ$ would make $|\mathbf{V}_{PD}^T \mathbf{V}_{PD}| = 0$, which indicates that the pseudo-inverse matrix of $\mathbf{V}_{PD}$ is incalculable, and the localization method can not work properly. Moreover, Fig. 8 shows that the three curves of the positioning error all have relatively low values at $\xi = 54.73^\circ$. This indicates that the optimal $\mathbf{V}_{PD}$ in (86) is a good choice to suppress the



(a) The positioning error of EGC derived in (46).
 $e_{ave} = 0.0623$ m. $e_{var} = 2.24 \times 10^{-4}$ m².



(b) The positioning error of EGC obtained with Monte Carlo simulations.



(c) The ratio of subfigure (b) and subfigure (a).

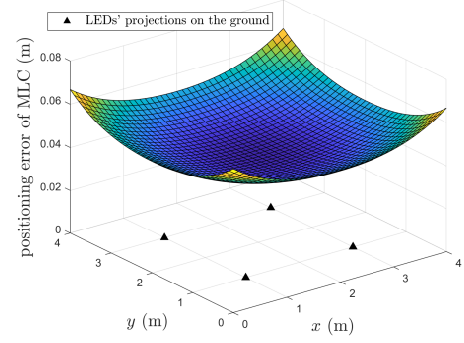
Fig. 5. The demonstration of EGC method's positioning error obtained with (46) and Monte Carlo simulations. Their ratios are plotted in subfigure (c) to verify the theoretical result in (46). Four LEDs are set at $[1 \text{ m}, 1 \text{ m}, 4 \text{ m}]^T$, $[1 \text{ m}, 3 \text{ m}, 4 \text{ m}]^T$, $[3 \text{ m}, 1 \text{ m}, 4 \text{ m}]^T$ and $[3 \text{ m}, 3 \text{ m}, 4 \text{ m}]^T$. The AOA estimator is set at $[x, y, 0 \text{ m}]^T$.

positioning error.

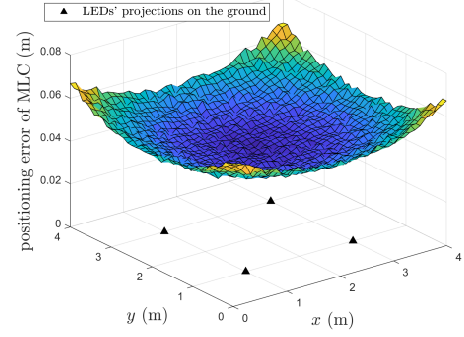
E. Accuracy of the Theoretical Error Expressions With Different Noise Levels

In this section, we would like to investigate whether the theoretical errors derived in (14), (36), (46) and (69) are accurate when the noise level changes. For testing, the indoor environment is set as $4 \times 4 \times 4$ m³, the AOA estimator is set at $[2 \text{ m}, 2 \text{ m}, 0 \text{ m}]^T$, while the LEDs for the single-LED, LED-select, EGC and MLC methods are set as those in Figs. 3-6. Besides, the standard deviation of the noise sample is magnified β times as

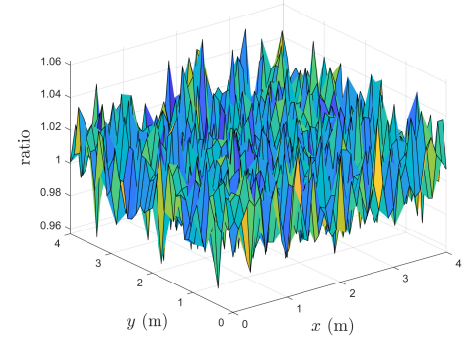
$$\sigma_{mag} = \beta \sigma_n = 5\beta \times 10^{-7} \text{ W.} \quad (87)$$



(a) The positioning error of MLC derived in (69).
 $e_{ave} = 0.0477$ m. $e_{var} = 3.06 \times 10^{-5}$ m².



(b) The positioning error of MLC obtained with Monte Carlo simulations.



(c) The ratio of subfigure (b) and subfigure (a).

Fig. 6. The demonstration of MLC method's positioning error obtained with (69) and Monte Carlo simulations. Their ratios are plotted in subfigure (c) to verify the theoretical result in (69). Four LEDs are set at $[1 \text{ m}, 1 \text{ m}, 4 \text{ m}]^T$, $[1 \text{ m}, 3 \text{ m}, 4 \text{ m}]^T$, $[3 \text{ m}, 1 \text{ m}, 4 \text{ m}]^T$ and $[3 \text{ m}, 3 \text{ m}, 4 \text{ m}]^T$. The AOA estimator is set at $[x, y, 0 \text{ m}]^T$.

The ratios of the theoretical and exact errors are plotted as the functions of β in Fig. 9. As shown in this figure, the four curves are close to 1 when $\beta \leq 10$, which verifies that the theoretical error expressions are accurate when the noise level is relatively low. However, when $\beta > 10$, the curves start to deviate from 1. This is because the high order small quantities in (15) and (70) are not negligible when the noise level increases.

VII. EXPERIMENTAL RESULTS

Fig. 10 demonstrates the experimental testbench of the size $2 \times 1 \times 1.03$ m³ for the proposed VLP methods, where

TABLE I
POSITIONING ERROR SAMPLES

Exact AOA estimator position (cm)	(45, 90, 0)	(45, 50, 0)	(45, 10, 0)	(95, 90, 0)	(95, 50, 0)	(95, 10, 0)	Average
Positioning with LED 1 (cm)	2.25	1.08	0.27	7.91	2.82	2.12	2.74
Positioning with LED 2 (cm)	9.94	6.54	1.91	4.76	0.48	0.53	4.03
Positioning with LED 3 (cm)	4.61	3.93	11.47	0.86	2.26	5.19	4.72
Positioning with LED 4 (cm)	1.13	1.00	4.80	3.07	2.94	8.25	3.53
Positioning with LED 5 (cm)	3.30	3.62	2.50	2.91	3.89	1.92	3.03
LED selection (cm)	1.13	1.00	2.50	0.86	2.26	1.92	1.61
EGC (cm)	3.43	2.14	3.41	1.50	1.04	2.59	2.35
MLC (cm)	0.49	0.98	1.24	0.60	1.49	0.94	0.96
Trilateration (cm)	6.67	4.34	9.54	4.58	2.32	6.79	5.71
Angulation (cm)	2.3	1.39	1.61	1.18	1.17	1.12	1.46

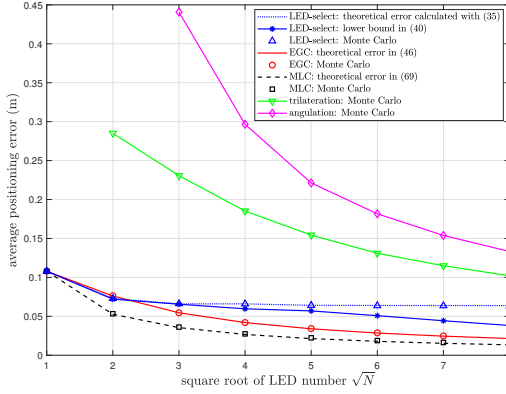


Fig. 7. The demonstration of the average positioning error of the testing points, which is plotted as the function of the LED number's square root $\sqrt{N}$.

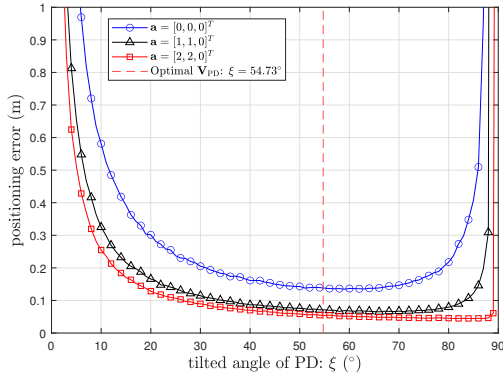


Fig. 8. The positioning error with the single-LED localization when the tilted angle of the PD changes. The results in Fig. 8 are obtained with Monte Carlo experiments. The LED is set at $[2 \text{ m}, 2 \text{ m}, 4 \text{ m}]^T$. The AOA estimator is set at $[0 \text{ m}, 0 \text{ m}, 0 \text{ m}]^T$, $[1 \text{ m}, 1 \text{ m}, 0 \text{ m}]^T$ or $[2 \text{ m}, 2 \text{ m}, 0 \text{ m}]^T$.

five LED arrays⁶ are installed on the ceiling, and the AOA estimator is equipped on the motorized XY stage to change its position for testing. Each LED array consists of sixteen LEDs to mimic an LED of 2 W, and the product number of each LED is OSRAM GW CSHPM1.CM. The AOA estimator contains eight tilted PDs, and their product numbers are Vishay BPW34. The sizes of the AOA estimator and the PD array are

⁶The LEDs can transmit in the random access or TDM mode to avoid collision, which can be realized with low-cost microcontroller units and low-complexity software [34].

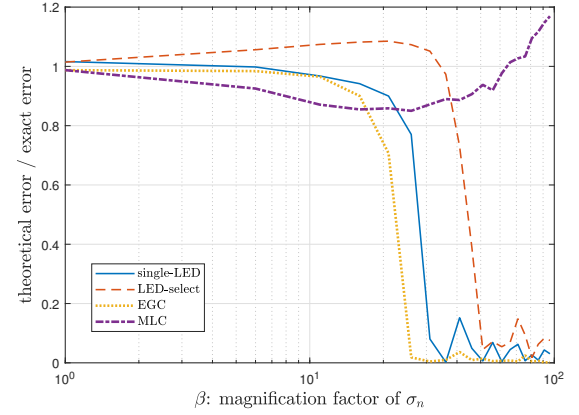


Fig. 9. The ratio of the theoretical error and the exact error (from Monte Carlo experiment) when the noise level changes. The AOA estimator is set at $[2 \text{ m}, 2 \text{ m}, 0 \text{ m}]^T$, while the LEDs for the single-LED, LED-select, EGC and MLC methods are set as those in Figs. 3-6.

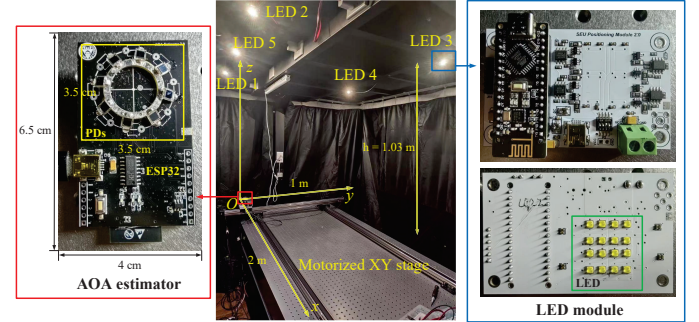


Fig. 10. Experimental testbench.

$6.5 \times 4 \text{ cm}^2$ and $3.5 \times 3.5 \text{ cm}^2$, respectively. The workflow of the testbench is shown in Fig. 11. The PD array in the AOA estimator receives the positioning signals transmitted from the LEDs, and each LED's signal is distinguished with the TDM scheme [34]. The received signals of the PDs are sampled by an analog-to-digital converter (ADC), whose product number is ADS70571RUGR. The sampled LED intensities are processed by a microcontroller unit (MCU), whose product number is ESP32-WROOM-32E, and then the AOAs of the LEDs are obtained. Finally, the AOA samples are sent to a laptop, which performs the proposed localization methods, and outputs the positioning results. The LED 1

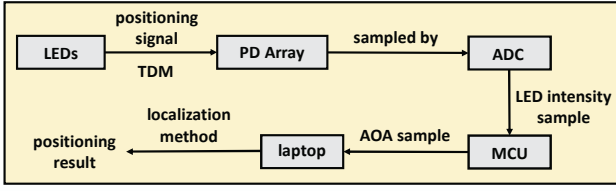


Fig. 11. Workflow of the experimental testbench in Fig. 10.

to LED 5 are placed at $[17.6 \text{ cm}, -16.5 \text{ cm}, 103 \text{ cm}]^T$, $[125.8 \text{ cm}, -15.7 \text{ cm}, 103 \text{ cm}]^T$, $[126.3 \text{ cm}, 95.6 \text{ cm}, 103 \text{ cm}]^T$, $[15.3 \text{ cm}, 96.7 \text{ cm}, 103 \text{ cm}]^T$ and $[71.7 \text{ cm}, -15.6 \text{ cm}, 103 \text{ cm}]^T$, respectively. Six testing points are selected for the AOA estimator, and the positioning errors of the single-LED, LED-select, EGC, MLC, classic trilateration and angulation are listed in Table I, which are the samples of one experiment at each testing point. At the right edge of Table I, the average positioning errors of the six testing points are calculated. These average positioning errors show that the proposed LED-select, EGC and MLC methods achieve better localization accuracies than that of the single-LED method. As shown in Table I, the positioning error for the single-LED method in (9) reaches the maximum when the receiver is far away. For example, the positioning error has a maximum of 11.47 cm when locating the AOA estimator at $[45 \text{ cm}, 10 \text{ cm}, 0 \text{ cm}]^T$ with the LED 3, which is because this testing point is the farthest from the LED 3. This result agrees with the analytical result in (14). Though the LED-select method cannot always choose the LED with the minimum positioning error due to the errors of the AOA samples, it can effectively abandon the localization results with big errors. The EGC and MLC methods are shown to improve the positioning accuracy, especially for the MLC method, which achieves the accuracy of the millimeter level. Last but not the least, our proposed LED-select, EGC and MLC methods have smaller average localization errors than that of the trilateration method, while the proposed MLC method has a smaller average localization error than that of the angulation method.

VIII. CONCLUSION

In this paper, we studied 2D VLP systems with multiple LEDs, and each of the LEDs was exploited to individually output a positioning result of an AOA estimator. Based on these single-LED positioning results, we proposed LED-select, EGC and MLC localization methods to fuse the AOA samples. The LED-select method chose the localization result obtained with the closest LED to the AOA estimator. The EGC method output the average of the results from all the LEDs. The MLC method maximized the likelihood function to achieve the highest accuracy. Closed-form positioning error expressions were derived for these methods. The proposed methods showed better localization accuracies than that of the single-LED VLP. Moreover, the accuracies of these methods could be improved by increasing the LED number, but the improvement was negligible when the LED number was sufficiently large. Finally, an experimental testbench was built to verify the

effectiveness of the proposed methods, and the millimeter-level accuracy was achieved.

APPENDIX A

If three events are denoted by A , B and C , the probability that they happen simultaneously is expressed as

$$\Pr(ABC) = \Pr(A)\Pr(B/A)\Pr(C/AB) \quad (88)$$

where $\Pr(B/A)$ denotes the probability that B happens under the condition that A has already happened. According to (88), $\Pr(\hat{\mathbf{a}}_{\text{sel}} = \hat{\mathbf{a}}_i)$ in (41) is expressed as

$$\begin{aligned} \Pr(\hat{\mathbf{a}}_{\text{sel}} = \hat{\mathbf{a}}_i) &= \Pr[\hat{r}_{i,3} = \max(\hat{r}_{1,3}, \dots, \hat{r}_{N,3})] \\ &= \Pr(\hat{r}_{i,3} > \hat{r}_{1,3}, \dots, \hat{r}_{i,3} > \hat{r}_{N,3}) \\ &= \prod_{j=1, j \neq i}^N \Pr(\hat{r}_{i,3} > \hat{r}_{j,3} / \cap_{k=1, k \neq i}^{j-1} \hat{r}_{i,3} > \hat{r}_{k,3}) \\ &= \prod_{j=1, j \neq i}^N \Pr(\hat{r}_{i,3} > \hat{r}_{j,3} / \underbrace{\hat{r}_{i,3} > \max\{\hat{r}_{k,3} | 1 \leq k \leq j-1, k \neq i\}}_{\hat{r}_{\max,j}}). \end{aligned} \quad (89)$$

If $\hat{r}_{j,3} \geq \hat{r}_{\max,j}$, we have $\Pr(\hat{r}_{i,3} > \hat{r}_{j,3} / \hat{r}_{i,3} > \hat{r}_{\max,j}) \geq \Pr(\hat{r}_{i,3} > \hat{r}_{j,3})$. Otherwise, we have $\Pr(\hat{r}_{i,3} > \hat{r}_{j,3} / \hat{r}_{i,3} > \hat{r}_{\max,j}) = 1 \geq \Pr(\hat{r}_{i,3} > \hat{r}_{j,3})$. Applying $\Pr(\hat{r}_{i,3} > \hat{r}_{j,3} / \hat{r}_{i,3} > \hat{r}_{\max,j}) \geq \Pr(\hat{r}_{i,3} > \hat{r}_{j,3})$ to (89), we obtain

$$\Pr(\hat{\mathbf{a}}_{\text{sel}} = \hat{\mathbf{a}}_i) \geq \prod_{j=1, j \neq i}^N \Pr(\hat{r}_{i,3} > \hat{r}_{j,3}), \quad (90)$$

which is exactly the lower bound in (41). If the bound in (90) is tight, the following condition should be satisfied:

$$\Pr(\hat{r}_{i,3} > \hat{r}_{j,3} / \hat{r}_{i,3} > \hat{r}_{\max,j}) = \Pr(\hat{r}_{i,3} > \hat{r}_{j,3}), \quad (91)$$

where $1 \leq j \leq N, j \neq i$. We can consider a special case that only the i th and the a th LEDs are close to the AOA estimator, while the other LEDs are far away. In this case, the b th ($\forall b \neq i, a$) LED's signal would become very weak when arriving at the AOA estimator, and thus $\hat{r}_{b,3}$ would be set to 0⁷. Since the LED is on the ceiling and the AOA estimator is on the ground, $\hat{r}_{i,3} > 0$ must be satisfied. Therefore, we have

$$\begin{aligned} \Pr(\hat{r}_{i,3} > \hat{r}_{b,3}) &= 1 \\ &= \Pr(\hat{r}_{i,3} > \hat{r}_{b,3} / \hat{r}_{i,3} > \max\{\hat{r}_{k,3} | 1 \leq k \leq b-1, k \neq i\}). \end{aligned} \quad (92)$$

Applying (92) to (89), we obtain

$$\begin{aligned} \Pr(\hat{\mathbf{a}}_{\text{sel}} = \hat{\mathbf{a}}_i) &= \Pr(\hat{r}_{i,3} > \hat{r}_{a,3} / \hat{r}_{i,3} > \max\{\hat{r}_{k,3} | 1 \leq k \leq a-1, k \neq i\}). \end{aligned} \quad (93)$$

Since $\max\{\hat{r}_{k,3} | 1 \leq k \leq a-1, k \neq i\} = 0$, $\hat{r}_{i,3} > \max\{\hat{r}_{k,3} | 1 \leq k \leq a-1, k \neq i\}$ must be satisfied. Thus, eq. (93) can be transformed as

$$\Pr(\hat{\mathbf{a}}_{\text{sel}} = \hat{\mathbf{a}}_i) = \Pr(\hat{r}_{i,3} > \hat{r}_{a,3}), \quad (94)$$

⁷In this case, the directional vector from the AOA estimator to the b th LED, i.e. $\mathbf{r}_b$, can be assumed parallel to the ground. Therefore, its z -axis entry $\hat{r}_{b,3}$ is 0.

which is reduced to the case of selecting an LED from the i th and the a th LEDs. Since $\Pr(\hat{r}_{i,3} > \hat{r}_{b,3}) = 1$ ($b \neq i, a$), eq. (94) is equivalent to

$$\Pr(\hat{\mathbf{a}}_{\text{sel}} = \hat{\mathbf{a}}_i) = \prod_{j=1, j \neq i}^N \Pr(\hat{r}_{i,3} > \hat{r}_{j,3}). \quad (95)$$

Therefore, the bound in (41) would be tight when only two LEDs are close to the AOA estimator while the other LEDs are far away.

APPENDIX B

Based on the estimated directional vector $\hat{\mathbf{r}}_i$ and the Pythagorean theorem [35, Appendix A], the distance from the AOA estimator to the i th LED can be estimated as

$$\hat{d}_i = \sqrt{\left(\frac{h}{\hat{r}_{i,3}} \hat{r}_{i,1}\right)^2 + \left(\frac{h}{\hat{r}_{i,3}} \hat{r}_{i,2}\right)^2 + h^2} \quad (96)$$

where $\hat{\mathbf{r}}_i = [\hat{r}_{i,1}, \hat{r}_{i,2}, \hat{r}_{i,3}]^T$. Since the coordinate of the i th LED is $\mathbf{t}_i = [t_{i,1}, t_{i,2}, h]^T$, we obtain

$$(\hat{a}_1 - t_{i,1})^2 + (\hat{a}_2 - t_{i,2})^2 + (0 - h)^2 = \hat{d}_i^2 \quad (97)$$

where $\hat{\mathbf{a}} = [\hat{a}_1, \hat{a}_2, 0]^T$ denotes the estimated position of the AOA estimator. Applying (96) to (97), we have

$$(\hat{a}_1 - t_{i,1})^2 + (\hat{a}_2 - t_{i,2})^2 = \underbrace{\left(\frac{h}{\hat{r}_{i,3}} \hat{r}_{i,1}\right)^2 + \left(\frac{h}{\hat{r}_{i,3}} \hat{r}_{i,2}\right)^2}_{d_{2D,i}^2}. \quad (98)$$

Substituting $i = 1, 2, \dots, N$ into (98), we can obtain the following N equations:

$$\begin{cases} (\hat{a}_1 - t_{1,1})^2 + (\hat{a}_2 - t_{1,2})^2 = d_{2D,1}^2, \\ \vdots \\ (\hat{a}_1 - t_{N,1})^2 + (\hat{a}_2 - t_{N,2})^2 = d_{2D,N}^2, \end{cases} \quad (99)$$

whose solution of $(\hat{a}_1, \hat{a}_2)$ is the result of the trilateration method, which can be obtained with the method in [36, Sec. IV-B].

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